NeoSignal

AI Intelligence Daily · Multi-Source · Cross-Verified

15 October 2026

353 verified · 830 raw · 5 sources

353	4	349	0
TOTAL	VERIFIED	CONFIRMED	EMERGING

Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

VERIFIED	**0.80 - 1.00**	3+ independent sources including media
CONFIRMED	**0.50 - 0.79**	2 sources or 1 high-quality media
EMERGING	**0.25 - 0.49**	Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	331	Media
The Verge AI	10	Media
TechCrunch AI	8	Media
HackerNews	7	Community
MIT Tech Review	1	Media

VERIFIED INTELLIGENCE — 4 stories

VERI

YouTube sticks new labels on AI-generated songs

The Verge AI +1 source 80%

YouTube is adding a "Made with AI" label to tracks that were "altered or fully generated" using AI, with a similar approach to its existing process for applying AI labels to videos. It doesn't sound like this will apply to any and all use of AI tools in making music, but...

https://www.theverge.com/ai-artificial-intelligence/1011369/youtube-music-ai-label

VERI

Governing Copyright Responsibly in the Era of Generative AI

ArXiv CS.AI +1 source 80%

arXiv:2610.17002v1 Announce Type: cross Abstract: Large language models are trained on collections that may include scholarly works, while authors generally lack work-level information about whether and how their works were used. This paper asks how copyright governance frames...

https://arxiv.org/abs/2610.17002

VERI

Comparing GenAI to Machine Learning and Manual Review for eDiscovery

ArXiv CS.AI +1 source 80%

arXiv:2610.16925v1 Announce Type: cross Abstract: Keeling et al. [Keeling2026Generative] report on a case study recently conducted by Redgrave LLP in which they assert that Relativity's Generative AI Review tool (aiR) achieved 88% recall while Relativity Assisted Learning (RAL),...

https://arxiv.org/abs/2610.16925

VERI

MemTrial: Learning When to Trust Memory in LLM Portfolio Agents

ArXiv CS.AI +1 source 80%

arXiv:2610.11732v2 Announce Type: replace Abstract: Large language model (LLM) agents for portfolio management learn from experience: they credit each experience in their memory with the outcome of the decisions that used it. In financial markets, however, this outcome mostly...

https://arxiv.org/abs/2610.11732

CONFIRMED SIGNALS — 349 stories

CONF

Show HN: Building an LLM from Scratch in TypeScript

HackerNews 50% HN 32

I wanted to understand transformers at a level where nothing is magic, so I'm building a ChatGPT-style model from first principles in TypeScript. Why not Python? I've found that reimplementing concepts in a different language forces me to grasp the nuance of the implementation....

https://book.humanexperiences.io/

CONF

Show HN: Jevcrawl – A web crawler for AI agents with no LLM in the crawl loop

HackerNews 50% HN 21

No summary — see link for full article.

https://github.com/RayenMlayeh/jevcrawl

CONF

Show HN: Ask a technical question, get an animated explainer you can interrupt

HackerNews 50% HN 10

No summary — see link for full article.

https://openlume.com/examples/3f65f6ce-87bf-4c81-a34b-3a1ec82f0ed1

CONF

Show HN: Wonderhop – A daily game: guess what most people say, or what none do

HackerNews 50% HN 10

Hello HN! Wonderhop is a small browser game with no account needed to play, everyday there are 4 rounds of 7 quesitons like "Name a food that a lot of people hate" The goal depends on the mode: Either gives the answer most people would give, the one no one would give, the oldest...

https://wonderhop.io/en/

CONF

Show HN: Chiplab - Give your AI agent access to virtual hardware

HackerNews 50% HN 7

No summary — see link for full article.

https://veecle.ai/chiplab

0 CONF

Show HN: MLX Direct Decider, Mac local decider without fine-tuning

HackerNews 50% HN 3

Hi HN, I built this open source project to enable the existing MLX model on your Mac to be used as a decider, without additional fine-tuning.

https://github.com/ItsOdeLeo/MLX-Direct-Decider

1 CONF

Show HN: An Email-to-iMessage Gateway

HackerNews 50% HN 2

Mobile carriers have long supported email-to-SMS functionality, but in the United States, the big three (Verizon, T-Mobile, and AT&T) have deprecated these services. So a crop of commercial email-to-SMS services have sprung up, and one of my SaaS tools is exactly that. But...

https://news.ycombinator.com/item?id=50088290

2 CONF

Ricursive Intelligence is betting on AI-designed chips to help solve the compute crisis

TechCrunch AI 50%

The founders detailed how their startup is developing AI that can design chips, learn from each iteration, and improve subsequent designs, all with the goal of automating and accelerating the chip design process.

https://techcrunch.com/2026/10/14/ricursive-intelligence-is-betting-on-ai-designed-chip...

3 CONF

Gudea leverages AI to help organizations prevent social media crises before they erupt

TechCrunch AI 50%

Every day, conversations unfold across social media and websites around the world. A seemingly insignificant post—whether it involves AI-generated misinformation, hate speech, or a damaging rumor—can quickly snowball into a much larger public controversy. Gudea, a company that...

https://techcrunch.com/2026/10/14/gudea-leverages-ai-to-help-organizations-prevent-soci...

4 CONF

Apate sends AI agents undercover to get scammed so they can gather fraud intelligence

TechCrunch AI 50%

Most companies use AI agents to ship products faster or speed up workflows. Apate.ai sends them undercover to get scammed.

https://techcrunch.com/2026/10/14/apate-sends-ai-agents-undercover-to-get-scammed-so-th...

5 CONF

Deezer will soon let you filter out AI-generated music

TechCrunch AI 50%

Deezer said that the new toggle will be rolled out by the end of the year.

https://techcrunch.com/2026/10/14/deezer-will-soon-let-you-filter-out-ai-generated-music/

6 CONF

Zenzap raises $44M for its AI chat app that aims to replace WhatsApp in the workplace

TechCrunch AI 50%

Zenzap is used by organizations including Starbucks and Taco Bell.

https://techcrunch.com/2026/10/14/zenzap-raises-44m-for-its-ai-chat-app-that-aims-to-re...

7 CONF

Floqer’s AI agents research sales prospects so teams don’t have to

TechCrunch AI 50%

The problem Floqer is trying to solve is familiar to anyone who has worked in sales or revenue operations.

https://techcrunch.com/2026/10/14/floqers-ai-agents-help-b2b-revenue-teams-research-cus...

8 CONF

Scimplify raises $25M to challenge China’s grip on critical materials with AI

TechCrunch AI 50%

Scimplify's annualized revenue run rate has tripled to $200 million as it eyes U.S. expansion.

https://techcrunch.com/2026/10/14/scimplify-raises-25m-to-challenge-chinas-grip-on-crit...

9 CONF

McKinsey is making job applicants spar with a fake AI client

The Verge AI 50%

Consulting firm McKinsey & Company is going to ask some entry-level candidates to have a 10-minute conversation with an AI simulation of a client as part of the interview process, according to The Wall Street Journal. The AI simulation "aims to reflect 80% of the scenarios...

https://www.theverge.com/news/1011654/mckinsey-ai-client-interview

0 CONF

Eight states sue OpenAI, saying ChatGPT isn’t safe for kids

The Verge AI 50%

OpenAI is facing a new wave of lawsuits from eight US states that accuse the company of harming children and misleading customers about the safety of ChatGPT, as reported earlier by Reuters. The lawsuits, filed separately by Iowa, Kansas, Nebraska, West Virginia, Arkansas,...

https://www.theverge.com/ai-artificial-intelligence/1011500/openai-state-lawsuits-chatg...

1 CONF

The AI race is back on

The Verge AI 50%

Last week, Anthropic and OpenAI looked like they had the AI race to themselves, pulling ahead with increasingly powerful models that left their rivals in the dust. Google seemed hopelessly far behind, while Meta, despite spending billions to revive its floundering AI ambitions,...

https://www.theverge.com/ai-artificial-intelligence/1011311/the-ai-race-is-back-on

2 CONF

Now that Character.AI is going bankrupt, what happens to all of its users’ data?

The Verge AI 50%

New York Attorney General Letitia James revealed a years-long multi-state investigation against Character.AI as they now seek to require protections on consumer data sold through its bankruptcy proceeding. James is among 40 bipartisan AGs who filed a notice of interest in...

https://www.theverge.com/policy/1011417/new-york-ag-characterai-bankruptcy

3 CONF

My AI doppelganger tells me what to eat

The Verge AI 50%

This is Optimizer, a weekly newsletter sent from Verge senior reviewer Victoria Song that dissects and discusses the latest gizmos and potions that swear they're going to change your life. Opt in for Optimizer here. For the past 100 days or so, my mornings have had a new rhythm....

https://www.theverge.com/column/1011264/digital-twin-metabolism-ai-twin-health

4 CONF

OpenAI claims an ex-employee took its product plans and presented them to Apple

The Verge AI 50%

OpenAI claims a former employee took more than 100 files related to its product plans before joining Apple. In a document filed on Wednesday, OpenAI accuses the unnamed employee of sending this information to a personal account and then presenting it to Apple leadership,...

https://www.theverge.com/ai-artificial-intelligence/1011276/openai-employee-unreleased-...

5 CONF

Square’s new kiosk lets you talk to AI for help with your order

The Verge AI 50%

Square is launching new hardware: a big kiosk for ordering things that has a touchscreen and lets you talk to an AI. It's called the Square Kiosk. Square says the Kiosk, which has a large pill shape, has an "agentic, conversational voice layer" that "fully understands a seller's...

https://www.theverge.com/tech/1009086/square-kiosk

6 CONF

Artificiality's First Exam: An LLM-Referenced C-test for Supraintelligence

ArXiv CS.AI 50%

arXiv:2610.14005v1 Announce Type: new Abstract: Under the accelerated pace of AI development, any benchmark whose difficulty is calibrated with humans or current AI systems becomes circular, or saturates soon. How can we then evaluate machines that may become more capable than...

https://arxiv.org/abs/2610.14005

7 CONF

ClawLoop: Self-Improving Agents via Harness-Native Training Loop

ArXiv CS.AI 50%

arXiv:2610.14118v1 Announce Type: new Abstract: The rapid advancement of Large Language Models (LLMs) has driven their deployment toward general-purpose harness environments, where LLM-based agents operate in sandbox workspaces and interact directly with file systems and shells....

https://arxiv.org/abs/2610.14118

8 CONF

The belief-action gap in language model agents

ArXiv CS.AI 50%

arXiv:2610.17444v1 Announce Type: cross Abstract: Language model (LM) agents can fail for different reasons, requiring different explanations and interventions. We study how state beliefs, action values, and action preferences interact as LM agents reason and commit to decisions...

https://arxiv.org/abs/2610.17444

9 CONF

Long-Horizon-Terminal-Bench 2.0: Automating Diverse Task Construction and Measuring Sustained Agent Progress

ArXiv CS.AI 50%

arXiv:2610.14146v1 Announce Type: new Abstract: Long-horizon terminal benchmarks measure whether an agent can sustain progress through extended workflows with process rewards, not only whether its final state passes a binary check. We present Long-Horizon-Terminal-Bench 2.0...

https://arxiv.org/abs/2610.14146

0 CONF

InCLAD: A Continual Learning Benchmark for Industrial Visual Anomaly Detection

ArXiv CS.AI 50%

arXiv:2610.15032v1 Announce Type: cross Abstract: Industrial anomaly detection benchmarks have advanced rapidly, yet they remain largely static and therefore underrepresent deployment settings in which inspection systems must be extended from one industrial component to another...

https://arxiv.org/abs/2610.15032

1 CONF

Unconstrained CTI-REALM with Open-Weights

ArXiv CS.AI 50%

arXiv:2610.14180v1 Announce Type: new Abstract: Organisations that want language-model agents for security work face a choice between open-weight models, which they can run and audit themselves, and far more expensive closed frontier models. We study that choice on CTI-REALM...

https://arxiv.org/abs/2610.14180

2 CONF

Monitoring latent reasoning in looped language models

ArXiv CS.AI 50%

arXiv:2610.18127v1 Announce Type: new Abstract: Looped language models (LMs) reuse one block of layers several times per token generation, promising more latent computation at no additional storage cost. Despite their rising popularity, the analysis of looped LMs is still in its...

https://arxiv.org/abs/2610.18127

3 CONF

Which Metadata Matters for LLM Tool Selection in MCP? Evidence from Controlled Factor-Level Interventions

ArXiv CS.AI 50%

arXiv:2610.14334v1 Announce Type: new Abstract: In Model Context Protocol (MCP) systems, LLMs select tools from metadata rather than from their implementations. Names and descriptions summarize what a tool does, while the parameter schema specifies the fields a tool accepts and...

https://arxiv.org/abs/2610.14334

4 CONF

Actionable Meta-Requirements for AI Robustness Auditing

ArXiv CS.AI 50%

arXiv:2610.14356v1 Announce Type: new Abstract: AI-enabled systems must remain dependable when the conditions under which they operate change (e.g., due to input perturbations, distribution shifts, or changes in external dependencies). This makes robustness auditing of such...

https://arxiv.org/abs/2610.14356

5 CONF

When Pleasing the User Compromises Care: AI Sycophancy in Caregiver-Mediated Healthcare

ArXiv CS.AI 50%

arXiv:2610.14360v1 Announce Type: new Abstract: Large language model (LLM) alignment often assumes that the user interacting with a model is also the primary beneficiary of its behavior. Yet in many consequential settings, one person directs the system while another bears the...

https://arxiv.org/abs/2610.14360

6 CONF

ACED-Bench: Evaluating How LLM Agents Acquire and Act on Causal Evidence

ArXiv CS.AI 50%

arXiv:2610.14381v1 Announce Type: new Abstract: Large language models are increasingly used as scientific assistants, yet we still lack reliable ways to evaluate whether they can investigate a system rather than merely answer a prompt. A useful evaluation must let agents choose...

https://arxiv.org/abs/2610.14381

7 CONF

ANASSA: From High-Level Framework to Low-Level Architecture for Spatial Agentic AI

ArXiv CS.AI 50%

arXiv:2610.14382v1 Announce Type: new Abstract: Agentic AI systems operating over geospatial data require more than autonomous reasoning and tool use; they must also support explicit orchestration, spatial validation, governance, provenance, failure handling, and human decision...

https://arxiv.org/abs/2610.14382

8 CONF

Measuring the Inference Energy Efficiency of Open-Weight LLMs across Benchmarks, Reasoning Modes, and Hardware

ArXiv CS.AI 50%

arXiv:2610.14389v1 Announce Type: new Abstract: Benchmark scores tell us what an LLM can do, but not how much energy is required to deliver that capability. We directly measure this trade-off for 86 open-weight model checkpoints, evaluated on seven benchmarks and six hardware...

https://arxiv.org/abs/2610.14389

9 CONF

Phantom Merge: When Your Large Language Model Agents Pick One but Tell You About Another

ArXiv CS.AI 50%

arXiv:2610.16141v1 Announce Type: new Abstract: LLM agents increasingly operate over multiple entities, where reliable response generation requires not only retrieving relevant evidence but also preserving its association with the correct entity. We identify Phantom Merge (PM),...

https://arxiv.org/abs/2610.16141

10 CONF

Frustratingly Easy Agentic RAG via Portable Skills: A Case Study in Customer Support

ArXiv CS.AI 50%

arXiv:2610.14428v1 Announce Type: new Abstract: Enterprise workflow automation typically requires retrieving a complete and accurate set of relevant information from large operational document corpora (SOPs, policy documents) given a specific scenario or request. In customer...

https://arxiv.org/abs/2610.14428

1 CONF

HiPPT: Physiology-Grounded Wearable PPG Representation Learning via Multi-Scope Text Alignment

ArXiv CS.AI 50%

arXiv:2610.14489v1 Announce Type: new Abstract: Reusable wearable PPG representations should capture local pulse morphology and broader physiological regularities. Existing pretraining objectives, however, usually apply supervision at the recording or pair level, overlooking...

https://arxiv.org/abs/2610.14489

2 CONF

Reels-RSI: Program-Level Self-Improvement for Code-Rendered Explainer Videos

ArXiv CS.AI 50%

arXiv:2610.14495v1 Announce Type: new Abstract: Explainer videos -- a worked geometry problem, why the sky is blue, how a dynasty fell -- are usually one more target for pixel video generation. We argue they are a different kind of artifact: a program (script, storyboard and...

https://arxiv.org/abs/2610.14495

3 CONF

Are Multimodal Benchmarks Really Useful? Item-Level Multimodal Benchmark Diagnosis via Structure-Response Co-Calibration

ArXiv CS.AI 50%

arXiv:2610.14542v1 Announce Type: new Abstract: Multimodal benchmarks are central to evaluating whether models can reason over information from multiple modalities. However, existing literature has revealed that some benchmarks may contain shortcut items that can be answered...

https://arxiv.org/abs/2610.14542

4 CONF

Iterate on an LLM, Serve on an SLM

ArXiv CS.AI 50%

arXiv:2610.14576v1 Announce Type: new Abstract: As frontier LLMs gain more capabilities, the cost to run them also goes up. At the same time, small language models (SLMs) also gain more capabilities but have largely been able to keep their cost stable, offering 5-40 times...

https://arxiv.org/abs/2610.14576

5 CONF

Parason: Revealing Subtask and Trial Parallelism in LLM Reasoning

ArXiv CS.AI 50%

arXiv:2608.24658v2 Announce Type: replace Abstract: Scaling test-time reasoning has substantially improved the problem-solving ability of large language models (LLMs), but standard autoregressive decoding still executes long reasoning traces sequentially, creating severe latency...

https://arxiv.org/abs/2608.24658

6 CONF

FETALFusion: Resolution-Aware State-Space Modeling for Multi-Domain Fetal Head Segmentation

ArXiv CS.AI 50%

arXiv:2610.17951v1 Announce Type: cross Abstract: Head circumference (HC) measured from 2D ultrasound is a routine marker of fetal growth, and automating it needs segmentations whose boundaries hold up on individual scans, not only on average. We present FETALFusion, a dual-path...

https://arxiv.org/abs/2610.17951

7 CONF

RT-J: Large-Scale Pretraining of Relational Transformers for Context-Efficient Predictions

ArXiv CS.AI 50%

arXiv:2610.17867v2 Announce Type: cross Abstract: Foundation models for tables and relational databases are increasingly used for predictive tasks across various scientific and industrial domains. However, adapting existing models to unseen datasets requires tens of thousands of...

https://arxiv.org/abs/2610.17867

8 CONF

Diff Diffusion: Predicting Edit Paths with Diffusion Models for Efficient Code Editing

ArXiv CS.AI 50%

arXiv:2610.17831v1 Announce Type: cross Abstract: Diffusion large language models (dLLMs) are appealing for code generation because they can update many tokens in parallel, yet many are optimized for sequence synthesis rather than edit prediction. This setting is poorly matched...

https://arxiv.org/abs/2610.17831

9 CONF

ECMR: Evidence-Aware Deliberation Control for Multi-Agent LLM Reasoning

ArXiv CS.AI 50%

arXiv:2610.17869v1 Announce Type: new Abstract: Multi-agent Large Language Model (LLM) systems typically commit to a deliberation protocol before inference. A fixed protocol cannot allocate computation differently to resolved and unresolved instances, and it may mistake repeated...

https://arxiv.org/abs/2610.17869

10 CONF

How Should a Camera-First AI Agent Work?

ArXiv CS.AI 50%

arXiv:2610.14772v2 Announce Type: new Abstract: Camera-first agents must act on the visual world as it unfolds. We argue that the central obstacle for future camera-first agents is not recognition ability - multimodal models (vision-language and live/streaming alike) generalize...

https://arxiv.org/abs/2610.14772

11 CONF

SAGE: Pushing 2-Bit Vector-Quantized LLMs toward Real-World Usability

ArXiv CS.AI 50%

arXiv:2610.14745v1 Announce Type: new Abstract: Extreme low-bit weight-only quantization is an effective approach to reducing the deployment costs of large language models. Below 3 bits, scalar quantization often suffers severe accuracy loss, while vector quantization (VQ)...

https://arxiv.org/abs/2610.14745

12 CONF

VH-GPO: Verified Handoff for LLM Agent Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2610.14000v1 Announce Type: cross Abstract: Group-relative reinforcement learning methods such as GRPO train large language model (LLM) agents by comparing rollouts for the same task. When every trajectory in a group fails with the same reward, relative advantages collapse...

https://arxiv.org/abs/2610.14000

3 CONF

GateSight: Recurrent Memory Anticipates Attention Memory for Query-Agnostic KV Eviction in Hybrid LLMs

ArXiv CS.AI 50%

arXiv:2610.14754v1 Announce Type: new Abstract: Hybrid language models interleave Gated DeltaNet (GDN) layers, whose recurrent memory has a fixed size, with full-attention layers, whose key-value (KV) cache grows with the context and dominates serving memory. In this paper, we...

https://arxiv.org/abs/2610.14754

4 CONF

CUSE: Fine-Grained Multimodal Unlearning with Counterfactual Guidance and On-Policy Self-Distillation

ArXiv CS.AI 50%

arXiv:2610.14763v1 Announce Type: new Abstract: Multimodal large language models (MLLMs) can reproduce sensitive, copyrighted, or unsafe information from their training data. Effective unlearning must remove target information while preserving non-target content, even within the...

https://arxiv.org/abs/2610.14763

5 CONF

Can Machine Learning Potentials Be More Accurate Than the Methods They Are Trained On?

ArXiv CS.AI 50%

arXiv:2610.18027v1 Announce Type: cross Abstract: Although machine learning potentials (MLPs) generalize across configurational and chemical compound spaces, their accuracy is commonly believed to be limited by their training methods, with improvements beyond this limit achieved...

https://arxiv.org/abs/2610.18027

6 CONF

A Reference Architecture for Integrating Agentic AI into Enterprise Software: Responsibilities, Interactions, and Variability

ArXiv CS.AI 50%

arXiv:2610.14818v1 Announce Type: new Abstract: Agentic AI moves generative AI applications toward systems that coordinate reasoning, enterprise context, model inference, and external actions. Integrating these capabilities into enterprise software introduces diverse...

https://arxiv.org/abs/2610.14818

7 CONF

AffectWorld: Benchmarking Emotion-Driven Future World Prediction

ArXiv CS.AI 50%

arXiv:2610.18305v1 Announce Type: cross Abstract: The world encompasses both physical environments and human actors. Predicting its evolution requires understanding both physical laws and how emotions shape human actions. However, existing world model benchmarks primarily assess...

https://arxiv.org/abs/2610.18305

8 CONF

WeatherBridge: Forecast-Aware Multi-Source Input Adaptation for Pretrained Weather Forecasting Models

ArXiv CS.AI 50%

arXiv:2610.14863v1 Announce Type: new Abstract: Data-driven weather forecasting models are commonly trained on ERA5, yet operational deployment must use timely initial states from physics-based numerical weather prediction (NWP) systems whose distributions differ substantially...

https://arxiv.org/abs/2610.14863

9 CONF

MARL-BattleGrounds: A Novel JAX-native Benchmark for Heterogeneous and Competitive Multi-agent RL

ArXiv CS.AI 50%

arXiv:2610.14883v1 Announce Type: new Abstract: We present MARL-BattleGrounds (MARL-BGs), an easy-to-use JAX-native benchmark for heterogeneous and competitive multi-agent reinforcement learning (MARL), designed to be sample-efficient, computationally efficient, and...

https://arxiv.org/abs/2610.14883

0 CONF

Double-Rollout Reinforcement Learning with Sampling-based Reward Estimation

ArXiv CS.AI 50%

arXiv:2610.16593v1 Announce Type: new Abstract: Reinforcement learning with verifiable rewards (RLVR) implemented with Group Relative Policy Optimization (GRPO) has driven substantial progress in training reasoning language models. However, verifiable outcome rewards provide an...

https://arxiv.org/abs/2610.16593

1 CONF

Single-Sample Decoding of a Two-Layer Sound-Word Canvas Model for Thai Klon Pad

ArXiv CS.AI 50%

arXiv:2610.14980v1 Announce Type: new Abstract: We present a generator of Thai klon pad verse, one stanza from a short topic, and evaluate it against human verse on held-out topics. The generator is a masked language model initialised from WangchanBERTa over a fixed canvas with...

https://arxiv.org/abs/2610.14980

2 CONF

Can Agents Truly Forget? Explore Machine Unlearning for LLM Agents

ArXiv CS.AI 50%

arXiv:2610.17659v1 Announce Type: new Abstract: Large language model (LLM) agents have shown strong capabilities in interactive problem solving, tool use, and long-horizon task execution. However, they may also acquire sensitive or private knowledge during training and expose it...

https://arxiv.org/abs/2610.17659

3 CONF

Belief Embedding Tree Search

ArXiv CS.AI 50%

arXiv:2610.15046v1 Announce Type: new Abstract: Beliefs in a Partially Observable Markov Decision Process (POMDP) are posterior distributions over the hidden state given the agent's history of actions and observations. They summarize everything the history says about the state,...

https://arxiv.org/abs/2610.15046

4 CONF

GesBench: When and Where Do Multimodal Language Models Perceive Social Meaning?

ArXiv CS.AI 50%

arXiv:2610.15129v1 Announce Type: new Abstract: Human social perception is incremental: we infer intent before an utterance is complete, revise our interpretation as new cues arrive, and attend to specific facial, gestural, or prosodic signals that shape social meaning. Current...

https://arxiv.org/abs/2610.15129

5 CONF

World Modeling through Spectral Alignment

ArXiv CS.AI 50%

arXiv:2610.15429v1 Announce Type: cross Abstract: How do we tell a world model what information to preserve? We study this question through the lens of spectral representation learning. We introduce SpecWM, a JEPA framework that explicitly specifies desired relationships between...

https://arxiv.org/abs/2610.15429

6 CONF

Unlocking Token Rewards via Training-Free Reward Attribution

ArXiv CS.AI 50%

arXiv:2610.18704v1 Announce Type: new Abstract: In this paper, we propose an extremely efficient, training-free method to extract token-level reward signals directly from an existing deep reward model. Our core idea is to attribute the overall process reward to individual tokens...

https://arxiv.org/abs/2610.18704

7 CONF

SocraticRepair: Planning-Guided Socratic Tutoring for Program Debugging and Misconception Repair

ArXiv CS.AI 50%

arXiv:2610.15204v1 Announce Type: new Abstract: Large language models are increasingly used in computer science education. However, the teaching abilities of even the most powerful LLMs are limited: they tend to reveal the solution prematurely and may lose focus on the learner's...

https://arxiv.org/abs/2610.15204

8 CONF

Running Fewer Loops Removes Refusal Behaviour in Recurrent-Depth Language Models

ArXiv CS.AI 50%

arXiv:2610.15218v1 Announce Type: new Abstract: A recurrent-depth language model reuses one block of weights several times in a row. The number of repetitions, which we call loops, is chosen when the model runs, not when it is trained, and running fewer is cheaper and faster, so...

https://arxiv.org/abs/2610.15218

9 CONF

Understanding and Reducing Misalignment in Persuasion Optimization

ArXiv CS.AI 50%

arXiv:2610.16070v1 Announce Type: new Abstract: Frontier models are now trained for complex multi-agent tasks, from running a company to engaging in negotiations. These tasks necessitate training models to be persuasive. But what does this optimization pressure mean for model...

https://arxiv.org/abs/2610.16070

0 CONF

OntoAR-HAR: Ontology-Guided Agentic RAG with Personalized Memory for Human Activity Recognition

ArXiv CS.AI 50%

arXiv:2610.15292v1 Announce Type: new Abstract: LLMs increasingly support sensor-based human activity recognition (HAR) by linking motion signals with activity semantics and physical knowledge. Retrieval-augmented generation (RAG) grounds decisions in sensor examples and domain...

https://arxiv.org/abs/2610.15292

1 CONF

Exploring Jailbreak Vulnerability Across Depth-Based Dynamic Inference in Large Language Models

ArXiv CS.AI 50%

arXiv:2610.15207v1 Announce Type: cross Abstract: Large Language Models (LLMs) have been shown to be computationally expensive and vulnerable to jailbreak attacks, in which adversarial prompts bypass safety alignment and induce harmful responses. To address this inefficiency, a...

https://arxiv.org/abs/2610.15207

2 CONF

MultiFlow: Vision-Driven Multimodal fMRI Encoding via Stimulus-Informed Source Flow Matching

ArXiv CS.AI 50%

arXiv:2610.15339v1 Announce Type: new Abstract: Current fMRI encoding models frame prediction as deterministic regression, yet identical stimuli evoke a distribution of neural responses (arising from physiological noise, hemodynamic variability, and cognitive fluctuations) that...

https://arxiv.org/abs/2610.15339

3 CONF

Fewer Bits, More Tries: Recovering Capability Under Memory Constraints

ArXiv CS.AI 50%

arXiv:2610.15372v1 Announce Type: new Abstract: Large language models are generally more capable than smaller ones, but their memory requirements can make local deployment difficult. We ask whether part of this capability gap can instead be recovered by running a smaller or more...

https://arxiv.org/abs/2610.15372

4 CONF

A Decade of PhotoMappers: A Longitudinal Study of Crowdsourced Disaster Photos Geolocalization with Vision-Language Models and Geospatial Reasoning

ArXiv CS.AI 50%

arXiv:2610.15403v1 Announce Type: new Abstract: Crowdsourced disaster imagery provides timely, ground-level observations when conventional disaster mapping is delayed or incomplete, yet its geographic value depends on where observations are produced, how they are interpreted,...

https://arxiv.org/abs/2610.15403

5 CONF

CoverSynth: Coverage-Guided Generation of Retrieval Evaluation Sets

ArXiv CS.AI 50%

arXiv:2610.15409v1 Announce Type: new Abstract: Enterprise agents need retrieval evaluation sets that expose domain-specific failures and guide developers towards efficient deployment tradeoffs. Neither expert-human-authored sets nor current synthetic generation methods provide...

https://arxiv.org/abs/2610.15409

6 CONF

SpecFit: Not All Draft Tokens Cost the Same in Offloaded MoE Inference

ArXiv CS.AI 50%

arXiv:2610.15417v1 Announce Type: new Abstract: Mixture-of-experts (MoE) models activate only a few experts per token, but when their expert weights are offloaded to host memory, host-to-device (H2D) expert transfers enter the critical path of decoding. Speculative decoding...

https://arxiv.org/abs/2610.15417

7 CONF

Hidden in Plain Sight: Advancing Harmful Content Understanding via Perception-Aware Adversarial Self-Play

ArXiv CS.AI 50%

arXiv:2610.15449v2 Announce Type: new Abstract: Despite significant advances in language understanding, Large Language Models (LLMs) can still miss meaning hidden in plain sight, failing to identify harmful intent in expressions readily understood by humans. Malicious users...

https://arxiv.org/abs/2610.15449

8 CONF

Imitation Holds a Key to Evolution: Localized Repair and Verified Skill Acquisition for Adaptive Agents

ArXiv CS.AI 50%

arXiv:2610.16125v1 Announce Type: new Abstract: An agent can retain successful trajectories without learning which part of an experience should repair a new discrepancy. We formulate passive imitation (PI) as an architectural loop that localizes a violated contract, extracts a...

https://arxiv.org/abs/2610.16125

9 CONF

When Skills Remain But Access Fails: Factorized Skill Memory for Continual Skill Internalization

ArXiv CS.AI 50%

arXiv:2610.15492v1 Announce Type: new Abstract: Skill internalization aims to convert externally specified procedural skills into persistent model capabilities, allowing language-model agents to execute them without retrieving or reinjecting skill descriptions at inference time....

https://arxiv.org/abs/2610.15492

0 CONF

VQEL: Enabling Self-Play in Emergent Language Games via Agent-Internal Vector Quantization

ArXiv CS.AI 50%

arXiv:2503.04940v3 Announce Type: replace-cross Abstract: Emergent Language (EL) focuses on the emergence of communication among artificial agents. Although symbolic communication channels more closely mirror the discrete nature of human language, learning such protocols remains...

https://arxiv.org/abs/2503.04940

1 CONF

Data Portability Revisited: Toward the Human-Centric, AI-Driven Data Ecosystems of Tomorrow

ArXiv CS.AI 50%

arXiv:2610.15503v1 Announce Type: new Abstract: This Article critically examines the contemporary regulatory framework and discourse surrounding data portability in the United States. Using recent regulatory developments in the European Union as an illustration, this Article...

https://arxiv.org/abs/2610.15503

2 CONF

TabGift: A Context-Efficient Foundation Model for Tabular Data Generation

ArXiv CS.AI 50%

arXiv:2610.16990v1 Announce Type: cross Abstract: Tabular data generation aims to synthesize realistic samples from limited observations while faithfully capturing the underlying data distribution. Tabular foundation models (TFMs), pretrained on diverse tabular tasks and adapted...

https://arxiv.org/abs/2610.16990

3 CONF

ScoringBench: A Benchmark for Evaluating Tabular Foundation Models with Proper Scoring Rules

ArXiv CS.AI 50%

arXiv:2603.29928v4 Announce Type: replace Abstract: Tabular foundation models such as TabPFN, TabICL, ExaOneTabular, and Nori already produce full predictive distributions, yet prevailing regression benchmarks evaluate them almost exclusively via point-estimate metrics such as...

https://arxiv.org/abs/2603.29928

4 CONF

Radioactive Feedback: Tracing Models Trained by Label-Only LLM Supervision

ArXiv CS.AI 50%

arXiv:2610.15685v1 Announce Type: new Abstract: Conventional distillation transfers capability through teacher-generated demonstrations. LLM-as-a-Judge knowledge distillation offers a distinct and increasingly practical alternative. In the label-only variant studied here, the...

https://arxiv.org/abs/2610.15685

5 CONF

Visible but Not Avoided: State-Dependent Re-Proposal of Rejected Candidates in Catalog-Based LLM Optimization Loops

ArXiv CS.AI 50%

arXiv:2610.15738v1 Announce Type: new Abstract: LLM optimization loops often show the proposer the candidates it has already tried and rejected, but leave it to the proposer to avoid them. We ask whether avoidance measured in a static probe predicts avoidance in the states an...

https://arxiv.org/abs/2610.15738

6 CONF

Structural Baselines Matter for Agent Failure Detection and Attribution

ArXiv CS.AI 50%

arXiv:2610.16661v1 Announce Type: new Abstract: An LLM agent using tools can fail silently. It stops before the task is done, raises no error, and answers fluently. Two kinds of monitor address this, one deciding whether a run failed and the other naming the step where failure...

https://arxiv.org/abs/2610.16661

7 CONF

AgentStripes: An Evidence-Graded Maturity Model for AI Agents

ArXiv CS.AI 50%

arXiv:2610.18603v1 Announce Type: new Abstract: Benchmarks report what an Artificial Intelligence (AI) agent can do on a task suite, but not whether one released version has been engineered with the controls that make it reliable, safe and observable, or how strongly that is...

https://arxiv.org/abs/2610.18603

8 CONF

Review Rarely Beats the Stronger Model: Separating Verification from Capability Transfer in LLM Pipelines

ArXiv CS.AI 50%

arXiv:2610.15897v2 Announce Type: cross Abstract: Qwen3-8B reviewing Llama-3.1-8B in math adds 33.2 points, yet Qwen3-8B alone is 6.4 points more accurate than the result. Reviewing another model's answer is usually judged by its gain over the generator. We show that this...

https://arxiv.org/abs/2610.15897

9 CONF

Comparative Study of Canvas- and Coordinate-Based Geometry-Conditioned Neural Networks for Real-Time Prediction of Indoor Environments with Varying Room Dimensions

ArXiv CS.AI 50%

arXiv:2610.15885v1 Announce Type: new Abstract: Computational fluid dynamics (CFD) is computationally expensive for iterative building design. Deep learning surrogates offer faster predictions, but generalization across varying room dimensions remains challenging. This study...

https://arxiv.org/abs/2610.15885

0 CONF

Multi-Horizon Explainable Queue Forecasting and Resource-Aware Decision Support for Intelligent Border Control Using LightGBM

ArXiv CS.AI 50%

arXiv:2610.15930v1 Announce Type: new Abstract: Border crossing points are stochastic service systems in which traffic demand, traveler categories, registration status, inspection requirements, staffing, and lane availability interact on multiple time scales. This paper extends...

https://arxiv.org/abs/2610.15930

1 CONF

The Anatomy of a Blind Spot: What an LLM Judge Must Be Shown, and How Much

ArXiv CS.AI 50%

arXiv:2610.15938v1 Announce Type: new Abstract: Evaluation of text generated by large language models (LLMs) is increasingly delegated to LLMs acting as judges, yet what such a judge must be shown remains unresolved. We treat a judge whose reports are acted on as a regulator, in...

https://arxiv.org/abs/2610.15938

2 CONF

Vision: "What Don't I Know About User's Preferences?" How Uncertainty Estimation Improves Proactive Assistant Alignment

ArXiv CS.AI 50%

arXiv:2610.15948v1 Announce Type: new Abstract: Proactive Smart Assistants (PSAs) increasingly use general knowledge, learned patterns from group-level data, patterns from individual user's data, and recent sensor observations to anticipate which action a user may prefer....

https://arxiv.org/abs/2610.15948

3 CONF

Vela: User Behavior as Supervision Sparks LLM Intelligence

ArXiv CS.AI 50%

arXiv:2610.14709v1 Announce Type: cross Abstract: Large language models are increasingly deployed in applications that act on users' long-term behavior histories, from personal assistants to recommendation and advertising systems. These applications demand the user dimension of...

https://arxiv.org/abs/2610.14709

4 CONF

Computational Reflective Equilibrium for Agentic AI: A Rawlsian Alignment Scaffold for Governance

ArXiv CS.AI 50%

arXiv:2610.16015v1 Announce Type: new Abstract: The rapid deployment of agentic AI has introduced complex challenges for the ethics and governance of AI systems. Agents built on large language models act through tools on behalf of people, and each deployment case brings its own...

https://arxiv.org/abs/2610.16015

5 CONF

Cross-Protocol Personality Assessment with LLMs: Aligning Behavioral Evidence with Human Ratings

ArXiv CS.AI 50%

arXiv:2610.16050v1 Announce Type: new Abstract: Personality assessment from asynchronous video interviews (AVIs) plays a crucial role in recruitment by enabling standardized and behaviorally grounded evaluation for job applicants. However, most of the previous works are confined...

https://arxiv.org/abs/2610.16050

6 CONF

The Sharp Directions Are Against You: Curvature Analysis of Activation Steering

ArXiv CS.AI 50%

arXiv:2610.16084v1 Announce Type: new Abstract: Activation steering adds a vector to a language model's internal representations at inference time. It is a lightweight alternative to fine-tuning for behavioral control. The standard construction, contrastive activation addition...

https://arxiv.org/abs/2610.16084

7 CONF

Constructed regional photovoltaic generation fields improve ultra-short-term distributed solar forecasting

ArXiv CS.AI 50%

arXiv:2610.16110v1 Announce Type: new Abstract: Ultra-short-term forecasting of densely distributed photovoltaic (PV) systems remains challenging because external meteorological and satellite products may not adequately resolve rapid local variations, while heterogeneous PV...

https://arxiv.org/abs/2610.16110

8 CONF

Large Language Model Assisted Transportation Planning through a Differentiable User Equilibrium Layer

ArXiv CS.AI 50%

arXiv:2610.16140v1 Announce Type: new Abstract: Transportation planning has traditionally required substantial expertise in network modeling, traffic assignment, and optimization. As a result, strategic analyses such as network design, capacity expansion, and model calibration...

https://arxiv.org/abs/2610.16140

9 CONF

SemTree: Judge-Aware Semantic Decision Trees for Inspectable and Repairable Rule Decisions

ArXiv CS.AI 50%

arXiv:2610.13999v1 Announce Type: cross Abstract: Rule decisions such as content moderation, advertising compliance, and eligibility checks combine several conditions, exemptions, and missing-information cases, and an error can be fixed only once the failing condition is known....

https://arxiv.org/abs/2610.13999

00 CONF

Uncovering Robustness Weaknesses in LLM Agents via Persona Evolution with Task-Integrity Constraints

ArXiv CS.AI 50%

arXiv:2610.16155v1 Announce Type: new Abstract: Interactive-agent evaluations commonly rely on standardized LLM-based user simulators that exhibit broadly cooperative behavior, potentially overlooking failures that arise under diverse but realistic user interactions. This can...

https://arxiv.org/abs/2610.16155

01 CONF

Get In, Decide, Prove: A Merchant-Side Readiness Benchmark for Agentic Commerce

ArXiv CS.AI 50%

arXiv:2610.17378v1 Announce Type: new Abstract: AI agents now browse and buy on people's behalf, and benchmarks measure how well the agents perform. Far less is known about the other side of the transaction: whether a merchant's website lets an agent in, publishes terms precise...

https://arxiv.org/abs/2610.17378

02 CONF

Learning to GUIDE the Group: Self-Learning LLM Facilitation Across Multi-Party Tasks

ArXiv CS.AI 50%

arXiv:2610.16231v1 Announce Type: new Abstract: As large language models are increasingly adopted as multi-party communication facilitators, there is a growing demand for agents capable of balancing diverse group dynamics across contexts. However, existing LLM facilitators are...

https://arxiv.org/abs/2610.16231

03 CONF

agent-risk: Risk Assessment for Changes to AI Agents

ArXiv CS.AI 50%

arXiv:2610.16280v1 Announce Type: new Abstract: Before changing a deployed AI agent (a stricter safety gate, a new model), organizations must approve the change, starting from existing logs. Extending classical risk analysis to partial records and partial changes to running...

https://arxiv.org/abs/2610.16280

04 CONF

Saliency-Guided Replay for Continual Learning in Deep Spiking Neural Networks

ArXiv CS.AI 50%

arXiv:2610.17998v1 Announce Type: cross Abstract: Spiking Neural Networks (SNNs) are promising for energy-efficient neuromorphic computing, but they still suffer from catastrophic forgetting when learning tasks sequentially. This work investigates the integration of Experience...

https://arxiv.org/abs/2610.17998

05 CONF

Narrative Paradoxes in LLM Inference: Shaping Reasoning Trajectories toward Misalignment

ArXiv CS.AI 50%

arXiv:2610.16312v1 Announce Type: new Abstract: Explicit reasoning in large language models (LLMs) is often associated with reliability, yet it can still pose risks when a model infers harmful goals under a benign context. We identify a distinct challenge: ensuring the safety of...

https://arxiv.org/abs/2610.16312

06 CONF

When a Conceptor Complement Becomes Gain: Operator Geometry and Behavioral Attribution in Language Models

ArXiv CS.AI 50%

arXiv:2610.16325v1 Announce Type: new Abstract: When a steering intervention changes a language model's behavior, which part of the intervention deserves the credit? We test this question for a target-centered, scaled conceptor complement by removing its fitted covariance filter...

https://arxiv.org/abs/2610.16325

07 CONF

Upstream Influence, Downstream Re-Anchoring in LLM Agent Chains

ArXiv CS.AI 50%

arXiv:2610.17723v1 Announce Type: new Abstract: People act on evaluative judgments produced by LLM agents, and in multi-agent systems those judgments pass through several agents before reaching a person. We show that a workflow's information architecture---source access, handoff...

https://arxiv.org/abs/2610.17723

08 CONF

RetroPlanner: Multi-Route Expansion and Multi-Objective Reasoning for LLM-Based Retrosynthesis Planning

ArXiv CS.AI 50%

arXiv:2610.16391v1 Announce Type: new Abstract: Multi-step retrosynthesis planning is often considered solved once a route to purchasable starting materials is found, but such a route should also be practical in real-world chemistry. Recent LLM-based planners formulate this...

https://arxiv.org/abs/2610.16391

09 CONF

PatTransBench: an ipc-stratified, multi-dimensional benchmark for patent machine translation evaluation

ArXiv CS.AI 50%

arXiv:2610.16538v1 Announce Type: cross Abstract: We present PatTransBench, a 12,000-item Chinese-English/Japanese/Korean patent machine translation (PatMT) benchmark with a four-dimensional evaluation protocol. The benchmark comprises 6,000 short-text (sentence/claim) and 6,000...

https://arxiv.org/abs/2610.16538

10 CONF

Compute-optimal data scaling for neural surrogates via multi-fidelity training

ArXiv CS.AI 50%

arXiv:2610.17221v1 Announce Type: cross Abstract: Neural surrogates for Partial Differential Equations (PDEs) are usually trained on synthetic data from numerical simulators. As models grow and scaling laws demand larger datasets, the cost of data generation often outweighs the...

https://arxiv.org/abs/2610.17221

11 CONF

Queue-Aware Token Communications: Bridging AI Representation and Wireless Service

ArXiv CS.AI 50%

arXiv:2610.16730v1 Announce Type: new Abstract: The rapid advancement of foundation models and edge intelligence is transforming wireless networks from data-oriented communication systems into artificial intelligence (AI)-native service platforms. Token communications (TokenCom)...

https://arxiv.org/abs/2610.16730

12 CONF

Same Illness, Different Voice: Restating the Patient's Account Narrows Narrative Anchoring in Clinical Language Models

ArXiv CS.AI 50%

arXiv:2607.27384v2 Announce Type: replace-cross Abstract: Clinical language models are evaluated on vignettes written in a clinician's voice, but patients describe the same illness in their own. We pair 1,000 MedQA cases with three retellings that carry the same audited original...

https://arxiv.org/abs/2607.27384

13 CONF

Acceptable Misalignment

ArXiv CS.AI 50%

arXiv:2610.16881v1 Announce Type: new Abstract: There is a specific challenge designers face in developing autonomous robots capable of open-ended learning. This challenge is how one can develop such systems in an ethically sound way when there are many unknowns during design...

https://arxiv.org/abs/2610.16881

14 CONF

Neural Bayesian Networks for Modular and Scalable Decision-Making Under Uncertainty

ArXiv CS.AI 50%

arXiv:2610.16902v1 Announce Type: new Abstract: Bayesian Networks (BNs) are established tools for letting agents reason under uncertainty in sequential decision-making. They split a joint distribution into local conditional mechanisms arranged on a directed acyclic graph, which...

https://arxiv.org/abs/2610.16902

15 CONF

Transparent Information Flow Modeling for Role-Aware AI Act Assessment

ArXiv CS.AI 50%

arXiv:2610.16916v1 Announce Type: new Abstract: The European Union's Artificial Intelligence Act introduces a structured framework for governing high-risk AI systems, emphasizing transparency, risk management, and role-specific accountability. However, current assessment...

https://arxiv.org/abs/2610.16916

16 CONF

PRAISE: A Reproducible Workflow for Non-Programmer Scientists to Implement Published AI Models

ArXiv CS.AI 50%

arXiv:2610.16980v1 Announce Type: new Abstract: Artificial Intelligence (AI) models are increasingly developed, published and used across scientific fields. Researchers, students and industry professionals who want to apply these models to their own research questions often lack...

https://arxiv.org/abs/2610.16980

17 CONF

Benchmarking Quantum Kernel SVMs for Credit Card Fraud Detection: Performance Realities Under NISQ Constraints

ArXiv CS.AI 50%

arXiv:2610.17027v1 Announce Type: new Abstract: Detecting credit card fraud is notoriously difficult because fraudulent charges make up less than 0.2% of total transactions. Quantum Machine Learning (QML), particularly Quantum Kernel Support Vector Machines (QSVM), promises...

https://arxiv.org/abs/2610.17027

18 CONF

SFT Leaves a Reproducible Signature in Activation Geometry: Calibrating Post-Training Interpretability Against Run-to-Run Noise

ArXiv CS.AI 50%

arXiv:2610.17093v1 Announce Type: new Abstract: A model diff credits post-training with whatever separates a base checkpoint from one post-trained descendant, presuming a rerun would land close by. On the sparse co-activation graph of Qwen3-1.7B it does not: two LoRA-SFT reruns...

https://arxiv.org/abs/2610.17093

19 CONF

When LLM Agents See a Conflict but Still Proceed: Cross-Domain Dependencies in Network Slicing

ArXiv CS.AI 50%

arXiv:2610.17118v1 Announce Type: new Abstract: Future AI-native networks are being designed to apply autonomous agents built on large language models (LLMs) in network management through intents. When intents from different domains meet on the same resource, an agent must...

https://arxiv.org/abs/2610.17118

20 CONF

EvoQuantLab: Training-Free Dual-Loop Evolution for Auditable Quantitative Research

ArXiv CS.AI 50%

arXiv:2610.17130v2 Announce Type: new Abstract: Automated alpha factor discovery is critical for modern quantitative research. While large language models (LLMs) have increasingly been integrated into quantitative workflows, existing agent-based approaches remain highly...

https://arxiv.org/abs/2610.17130

21 CONF

Learning Perceptual Geometry of Natural Odor Mixtures from LLM-Derived Descriptors

ArXiv CS.AI 50%

arXiv:2610.17164v1 Announce Type: new Abstract: The relationship between odor chemistry and perception remains incompletely understood. Many existing studies focus on the perception of single molecules, whereas natural odors contain tens to hundreds of components, adding extra...

https://arxiv.org/abs/2610.17164

22 CONF

AutoMARS: Automatic and Scalable Multi-Agent Large Language Model Framework for Supply Chain Risk Assessment

ArXiv CS.AI 50%

arXiv:2610.17206v1 Announce Type: new Abstract: We propose AutoMARS (Automatic Multi-Agent Risk Assessment), an agentic framework for automatic and scalable comparative risk assessment in supply chain networks. The framework enables end-to-end execution of analytical workflows...

https://arxiv.org/abs/2610.17206

23 CONF

Reliable Text-to-SQL in Resource-Constrained Settings with Small LLMs via Iterative Retry Workflows

ArXiv CS.AI 50%

arXiv:2610.17305v1 Announce Type: new Abstract: Large Language Models (LLMs) have recently improved Text-to-SQL generation, but their computational cost can hinder deployment in resource-constrained settings. This paper evaluates the Text-to-SQL capabilities of small and medium...

https://arxiv.org/abs/2610.17305

24 CONF

Budgeted KV-Cache Repair for Evolving Contexts

ArXiv CS.AI 50%

arXiv:2610.17338v1 Announce Type: new Abstract: Language-model agents revise their working context as information becomes obsolete, but deleting text does not erase its influence on cached computation. Later key-value (KV) states remain conditioned on the deleted content, while...

https://arxiv.org/abs/2610.17338

25 CONF

SchemaWeaver: Accelerating On-Device Mobile Agentic AI via Schema-Guided Parallel LLM Decoding

ArXiv CS.AI 50%

arXiv:2610.17436v1 Announce Type: new Abstract: Large language models (LLMs) on mobile devices can act as agents that call external software tools to complete complex tasks, but the major challenge of supporting on-device mobile agentic AI with LLMs is the long latency of LLM...

https://arxiv.org/abs/2610.17436

26 CONF

Large Language Models Extend Known Research But Do Not Create New Research Directions

ArXiv CS.AI 50%

arXiv:2610.17439v1 Announce Type: new Abstract: The success of autonomous research systems relies on the quality of the scientific hypotheses they generate. Recent studies report that human reviewers often rate LLM-generated ideas as more novel than those from human experts,...

https://arxiv.org/abs/2610.17439

27 CONF

Beyond Bias Scores: Answerability Confounds in Activation Steering with Gradient-Constrained Direction Design (GCD)

ArXiv CS.AI 50%

arXiv:2610.17641v1 Announce Type: new Abstract: Bias scores in language-model evaluations often conflate what a model prefers with whether it answers at all. We identify this confound in \textsc{BBQ}, whose ambiguous score factorizes into answerability and signed preference....

https://arxiv.org/abs/2610.17641

28 CONF

ProAS: A Progress-Aware Advantage Shaping Framework for Multi-Turn Agentic Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2610.18683v1 Announce Type: cross Abstract: Reinforcement learning (RL) has emerged as the primary paradigm for optimizing Large Language Model (LLM) agents, yet efficient critic-free algorithms (e.g., GRPO) suffer from a fundamental uniform credit assignment dilemma that...

https://arxiv.org/abs/2610.18683

29 CONF

DuplexAct: How Interactive Are Frontier Full-Duplex Spoken Language Models?

ArXiv CS.AI 50%

arXiv:2610.17738v1 Announce Type: new Abstract: End-to-end (E2E) full-duplex spoken language models (FD-SLMs) enable simutanous listening and speaking continously, requiring models to take appropriate interaction actions as conversations unfold. However, existing benchmarks lack...

https://arxiv.org/abs/2610.17738

30 CONF

Instructions Are Evidence, Not Objectives: Measuring Outcome Alignment of Agents

ArXiv CS.AI 50%

arXiv:2610.17757v1 Announce Type: new Abstract: Large language model agents can follow complex human instructions to solve real-world tasks. In practice, however, initial instructions are often incomplete. In our analysis of 100 user-agent trajectories from SWE-chat, 56\% of...

https://arxiv.org/abs/2610.17757

31 CONF

Can an Agent Learn to Predict Its Own Failure? Outcome-Grounded Confidence Calibration in LLM Agents

ArXiv CS.AI 50%

arXiv:2610.18614v1 Announce Type: new Abstract: Controlling an acting LLM agent means deciding, at each step of a long-horizon task, whether to trust its next action, abstain, or hand off to a human. That decision needs a per-step confidence signal predicting whether the current...

https://arxiv.org/abs/2610.18614

32 CONF

Learning Shortest Paths with Generative Flow Networks

ArXiv CS.AI 50%

arXiv:2603.01786v2 Announce Type: replace-cross Abstract: In this paper, we present a novel learning framework for finding shortest paths in graphs utilizing Generative Flow Networks (GFlowNets). First, we examine theoretical properties of GFlowNets in non-acyclic environments...

https://arxiv.org/abs/2603.01786

33 CONF

Beyond Questions: Evaluating the Stated Knowledge of LLMs

ArXiv CS.AI 50%

arXiv:2605.26937v3 Announce Type: replace-cross Abstract: Parametric knowledge in large language models (LLMs) is a cornerstone of their success, yet remains poorly understood. Existing knowledge benchmarks typically rely on predefined questions (e.g., "What is the birth date of...

https://arxiv.org/abs/2605.26937

34 CONF

ResearchFlow-Eval: Tracing the Execution Gap in End-to-End Multi-Claim Autoresearch

ArXiv CS.AI 50%

arXiv:2610.18088v2 Announce Type: new Abstract: Autoresearch systems increasingly undertake entire research projects. In practice, scientific research requires systems to pursue multiple interdependent claims through a continuous, multi-stage workflow and revise experiments as...

https://arxiv.org/abs/2610.18088

35 CONF

Mitigating Cross-Modal Hallucinations by Correcting Modality Misrouting in Omnimodal LLMs

ArXiv CS.AI 50%

arXiv:2610.18141v1 Announce Type: new Abstract: Omnimodal large language models (OLLMs) are susceptible to cross-modal hallucinations under audio-visual interference. We find that, for a substantial fraction of incorrect joint predictions, the question-relevant unimodal branch...

https://arxiv.org/abs/2610.18141

36 CONF

Neuro-Symbolic Graph Transformers for Hierarchical Intention Recognition from Multimodal Human Activity Sequences

ArXiv CS.AI 50%

arXiv:2610.18176v1 Announce Type: new Abstract: Proactive human-robot collaboration requires autonomous systems to infer not only immediate actions but also the underlying hierarchical intentions governing human behavior. Current deep learning approaches predominantly focus on...

https://arxiv.org/abs/2610.18176

37 CONF

Efficient Safety Guardrails through Latent Dual-Hypothesis Reasoning with On-Demand Trace Reconstruction

ArXiv CS.AI 50%

arXiv:2610.18304v1 Announce Type: new Abstract: Guardrail approaches that rely on reasoning-based large language models (LLM) improve safety decisions, but come with substantial inference overhead due to the explicit reasoning. We introduce LR-Guard (Latent Reasoning Guardrail),...

https://arxiv.org/abs/2610.18304

38 CONF

Thinker-WM: A Unified World Model with Shared Multimodal Representations for Embodied Intelligence

ArXiv CS.AI 50%

arXiv:2610.18447v2 Announce Type: new Abstract: World models have emerged as a promising paradigm for embodied AI, yet existing systems often remain fragmented in the modalities they model and the functions they support. In particular, visual dynamics, geometric structure, and...

https://arxiv.org/abs/2610.18447

39 CONF

Dream2Discern: Learning from Hard Latent Variants for Runtime Failure Prediction in Multi-Agent Systems

ArXiv CS.AI 50%

arXiv:2610.16071v1 Announce Type: cross Abstract: Reliable deployment of LLM-based multi-agent systems calls for runtime auditors that predict terminal outcomes from observed execution prefixes. These auditors typically learn from a limited corpus of trajectories with verified...

https://arxiv.org/abs/2610.16071

40 CONF

Fan Out or Relay? Ownership-Aware Routing for LLM Agents Querying Siloed Tables

ArXiv CS.AI 50%

arXiv:2610.18157v1 Announce Type: cross Abstract: Inside organizations, a question's data often sit with several owners, who query their own tables but never hand them over. An LLM coordinator can fan out, sending the question to every owner at once, or relay, querying owners in...

https://arxiv.org/abs/2610.18157

41 CONF

Rethinking Quality in Large Language Model Routing: Security-Conditioned Model-and-Budget Selection

ArXiv CS.AI 50%

arXiv:2610.18566v1 Announce Type: new Abstract: Large language model (LLM) routers reduce serving cost by choosing which model to call and how many output tokens to grant it for each query. Existing routers rank model-budget actions using a single quality metric. This does not...

https://arxiv.org/abs/2610.18566

42 CONF

Configuring RAG Systems: What Breaks, Where Originates, and How to Fix

ArXiv CS.AI 50%

arXiv:2610.18567v1 Announce Type: new Abstract: Retrieval-Augmented Generation (RAG) has emerged as a dominant paradigm for improving Large Language Model (LLM) by intertwining it with other components, such as a vector/graph database. Therefore, the resulting RAG systems are...

https://arxiv.org/abs/2610.18567

43 CONF

Are LLM Agents Behaviorally Coherent? A Latent Profiling Approach for Social Simulation

ArXiv CS.AI 50%

arXiv:2509.03736v3 Announce Type: replace Abstract: LLM agents are increasingly proposed as substitutes for human participants in social research, usually on the strength of survey responses that match those of the humans they represent. We argue that such output-level alignment...

https://arxiv.org/abs/2509.03736

44 CONF

Reading Right, Acting Wrong: Diagnosing Decision Failures of System One Models in Agent Loops

ArXiv CS.AI 50%

arXiv:2610.18579v1 Announce Type: new Abstract: System One models choose an agent's next action by scoring a supplied set of options in a single pass. This makes them fast and cheap, but when they choose wrongly there is nothing to inspect: did the model misread what the agent...

https://arxiv.org/abs/2610.18579

45 CONF

Compare to Correct: Pairwise-Guided Logit Adjustment for LLM Test-Time Scaling

ArXiv CS.AI 50%

arXiv:2610.18583v1 Announce Type: new Abstract: Sampling-based test-time scaling enhances LLM reasoning by aggregating multiple reasoning paths for the same query. Existing methods use answer frequencies or path confidence to compute candidate scores, which do not explicitly...

https://arxiv.org/abs/2610.18583

46 CONF

Beyond Answer Voting: Fine-Grained Correction of LLM Reasoning through Graph Alignment across Sampled Traces

ArXiv CS.AI 50%

arXiv:2610.18606v1 Announce Type: new Abstract: Large language models increasingly tackle tasks that require multi-step reasoning, where errors in intermediate claims can affect the final answer. Self-consistency (SC) improves reasoning by voting over sampled final answers, but...

https://arxiv.org/abs/2610.18606

47 CONF

Two Routes to Interoceptive Prioritisation: Dissociating Likelihood Precision and Preference Reweighting in Active Inference

ArXiv CS.AI 50%

arXiv:2610.18617v1 Announce Type: new Abstract: An embodied agent must prioritise among competing bodily needs. In active inference, it can respond to an urgent need by increasing the precision of information about that need or by increasing its preference for outcomes that...

https://arxiv.org/abs/2610.18617

48 CONF

To Align or Not To Align: Check Your COMPASS Before You Train

ArXiv CS.AI 50%

arXiv:2610.18638v1 Announce Type: new Abstract: Combining pretrained models is a common strategy for building AI systems, whether by aligning a vision encoder with a language model or merging fine-tuned checkpoints. However, finding compatible pairs typically requires costly...

https://arxiv.org/abs/2610.18638

49 CONF

BLADE: Auditing Information Use for Failure Attribution in LLM Multi-Agent Systems

ArXiv CS.AI 50%

arXiv:2610.18780v1 Announce Type: new Abstract: Large language model multi-agent systems extend individual agents' capabilities through collaboration across specialized roles, tools, and workflows. Yet collaboration makes failures harder to diagnose: one agent may misinterpret a...

https://arxiv.org/abs/2610.18780

50 CONF

EmoXpt: Analyzing Emotional Variances in Human Comments and LLM-Generated Responses

ArXiv CS.AI 50%

arXiv:2501.06597v1 Announce Type: cross Abstract: The widespread adoption of generative AI has generated diverse opinions, with individuals expressing both support and criticism of its applications. This study investigates the emotional dynamics surrounding generative AI by...

https://arxiv.org/abs/2501.06597

51 CONF

Dependency, Compression, and Synergy: A Unified Information-Theoretic View of Multimodal Learning

ArXiv CS.AI 50%

arXiv:2609.14421v1 Announce Type: cross Abstract: Recent advances in multimodal foundation models have intensified the need to understand how different modalities share, preserve, and complement information. Mutual Information (MI), the Information Bottleneck (IB), and Partial...

https://arxiv.org/abs/2609.14421

52 CONF

Coding LLMs Can Be Harnessed to Learn Interpretable Models from Data

ArXiv CS.AI 50%

arXiv:2610.14006v1 Announce Type: cross Abstract: Since compact, interpretable classifiers correspond to simple programs, it is natural to ask if LLM-based coding agents can be used to learn interpretable models from data. In this paper we present a simple harness that...

https://arxiv.org/abs/2610.14006

53 CONF

What Can Language Models Inherit from Solomonoff Induction? Limits, Attainable Properties, and What Served Conditionals Reveal

ArXiv CS.AI 50%

arXiv:2610.14007v1 Announce Type: cross Abstract: A familiar line of reasoning runs as follows: language models are flexible approximators trained under a strictly proper loss, so scale brings them closer to Solomonoff induction, and their predictive success is evidence of that...

https://arxiv.org/abs/2610.14007

54 CONF

When Log Parsing Removes What Matters: Detecting Failures in Distributed GPU Workloads

ArXiv CS.AI 50%

arXiv:2610.14009v1 Announce Type: cross Abstract: Large AI models increasingly rely on distributed GPU execution, yet it remains unclear whether established failure analysis methods transfer effectively to this setting. We introduce GF-Bench, a PyTorch-based framework and...

https://arxiv.org/abs/2610.14009

55 CONF

AeroChem: Closed-loop Physics-Informed State Space Modeling for Long-term Chemically-Reactive Air Quality Forecasting--Extended Version

ArXiv CS.AI 50%

arXiv:2610.16630v1 Announce Type: cross Abstract: Accurate long-term air quality forecasting is challenging due to long-range transport, meteorological variability, and nonlinear chemical reactions underlying secondary pollutant formation. Purely data-driven models often suffer...

https://arxiv.org/abs/2610.16630

56 CONF

Pick the Mixture, Not the Samples: Combinatorial Data Selection for LLMs

ArXiv CS.AI 50%

arXiv:2610.14028v1 Announce Type: cross Abstract: Data selection for LLM fine-tuning faces two facts about real training: data pools draw from many distinct domains, and evaluation spans diverse tasks. As a result, the same sample helps different tasks differently, and samples...

https://arxiv.org/abs/2610.14028

57 CONF

A CaLLM Judge is a Better Judge: Mitigating Bias in LLM-as-a-judge

ArXiv CS.AI 50%

arXiv:2610.18108v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly used as judges to predict which of two candidate answers a human evaluator would prefer, providing a scalable alternative when human evaluation is too costly. Unfortunately, LLM...

https://arxiv.org/abs/2610.18108

58 CONF

Rotate the Loop: Blockwise Conformality for Looped Transformers

ArXiv CS.AI 50%

arXiv:2610.14030v1 Announce Type: cross Abstract: Looped transformers reuse one shared block across many recurrent steps, so depth becomes a quantity that can be spent at runtime rather than a stack of distinct layers. Those extra steps help only if a write made on one recurrent...

https://arxiv.org/abs/2610.14030

59 CONF

The Future of AI-assisted Physics: Discovering, Verifying and Organizing Results via Formalization

ArXiv CS.AI 50%

arXiv:2610.14059v1 Announce Type: cross Abstract: Rapid advances in AI have opened a new era for physics, enabling unprecedented capabilities in research and discovery. However, these advances also bring challenges in verifying, trusting, and reusing the results they so rapidly...

https://arxiv.org/abs/2610.14059

60 CONF

Two-Stage Multi-modal Learning for Dynamic Audio-Visual Navigation: From Physical Alignment to Scene Aggregation

ArXiv CS.AI 50%

arXiv:2610.14077v1 Announce Type: cross Abstract: Embodied Audio-Visual Navigation (AVN) requires an agent to locate sound targets by jointly leveraging visual and auditory cues. While prior work has achieved notable success with static sound sources, real-world scenarios often...

https://arxiv.org/abs/2610.14077

61 CONF

SAGE: Efficient Protected Vision-Language Inference on Smart Glasses

ArXiv CS.AI 50%

arXiv:2610.14089v1 Announce Type: cross Abstract: Smart glasses are a promising platform for vision-language assistants, but private split inference remains expensive: protecting visual representations before transmission does not remove the cost of on-device vision encoding,...

https://arxiv.org/abs/2610.14089

62 CONF

Embodied Zero-Shot Task Planning via Hardware-Aware LLM-VLM Integration for Grounded Robotic Manipulation

ArXiv CS.AI 50%

arXiv:2610.14106v1 Announce Type: cross Abstract: Large language models can translate natural-language instructions into plausible manipulation plans, but robotic execution requires physically grounded decisions that respect robot embodiment, workspace geometry, reachability,...

https://arxiv.org/abs/2610.14106

63 CONF

This Is a Good Paper, Right? How Tag Questions Affect Sycophancy Across 106 Language Models

ArXiv CS.AI 50%

arXiv:2607.23976v2 Announce Type: replace-cross Abstract: Whether a language model agrees with a user depends on how the user asks. We measure this on 106 models with a grid of three wordings and three cues, on 20 personal decisions, counterbalanced over both options and scored...

https://arxiv.org/abs/2607.23976

64 CONF

Reinforcement Learning In Partially Verifiable Environments for Colon Cancer Treatment Planning

ArXiv CS.AI 50%

arXiv:2610.17426v1 Announce Type: cross Abstract: Clinical decision-making is a sequential process of information collection, clinical inference, and forecasting under partial observability, often without a uniquely correct treatment decision. Large language models (LLMs) have...

https://arxiv.org/abs/2610.17426

65 CONF

Deep learning from a control vantage point

ArXiv CS.AI 50%

arXiv:2610.14385v1 Announce Type: cross Abstract: This article offers an overview of recent developments in deep learning from a control vantage point. It reviews several classes of models including feedforward and residual neural networks, transformers, large-language models,...

https://arxiv.org/abs/2610.14385

66 CONF

RealityBench: Simulated Environments to Evaluate a Vision Language Model's Ability to Document Human Physical Work

ArXiv CS.AI 50%

arXiv:2610.14397v2 Announce Type: cross Abstract: Augmenting worker capabilities in the physical world through guidance has been a decades-long goal for AI agents. One key supporting capability is to understand what the worker is doing, seeing through the human's camera or smart...

https://arxiv.org/abs/2610.14397

67 CONF

Coarse-Graining Recursive Expert Computation: Continuation-Relative Interfaces and Failure Modes

ArXiv CS.AI 50%

arXiv:2610.14413v1 Announce Type: cross Abstract: Computational coarse-graining aims to replace a sequence of updates with a cheaper module while preserving task performance. We present a continuation-relative evaluation framework that assesses replacements through the...

https://arxiv.org/abs/2610.14413

68 CONF

Diffusion-Based Auctions for Multi-Winner Banner Ads

ArXiv CS.AI 50%

arXiv:2610.14473v1 Announce Type: cross Abstract: We propose a new framework for banner ad auctions, leveraging diffusion models to allow for multiple bidders to co-win a single banner ad opportunity, and in particular influence the ad shown on the banner depending on the...

https://arxiv.org/abs/2610.14473

69 CONF

RES-MR: Risk-Aware Reasoning for Explainable and Safe Medication Recommendation

ArXiv CS.AI 50%

arXiv:2610.14474v1 Announce Type: cross Abstract: Medication recommendations seek to deliver personalized drug combinations based on patients' clinical information, where ensuring transparent decision-making and personalized safety are of paramount importance. However, existing...

https://arxiv.org/abs/2610.14474

70 CONF

GeoMotionGPT: Geometry-Aligned Motion Understanding with Large Language Models

ArXiv CS.AI 50%

arXiv:2601.07632v5 Announce Type: replace-cross Abstract: Discrete motion tokenization has recently enabled Large Language Models (LLMs) to serve as versatile backbones for motion understanding and motion-language reasoning. However, existing pipelines typically decouple motion...

https://arxiv.org/abs/2601.07632

71 CONF

Compositional Concepts Erasing: Disentangling Object-Style via Domain Manifold Unlearning

ArXiv CS.AI 50%

arXiv:2610.14514v1 Announce Type: cross Abstract: Diffusion models pose significant risks regarding the generation of illicit content and copyright violations. This has catalyzed the development of machine unlearning to erase harmful concepts or copyrighted styles from a model's...

https://arxiv.org/abs/2610.14514

72 CONF

Who to Trust in Multi-Robot Systems: Joint Robot and Object Hypothesis Trust Inference via Neural-Symbolic Cross-Validation

ArXiv CS.AI 50%

arXiv:2610.14530v1 Announce Type: cross Abstract: Multi-robot systems (MRS) offer significant advantages over single-robot platforms by enabling information sharing and coordination for different tasks in complex environments. However, reliable cooperation critically depends on...

https://arxiv.org/abs/2610.14530

73 CONF

Active Neural Matter: A Finite-Window Response Framework for Physical Systems and Distributed AI Workflows

ArXiv CS.AI 50%

arXiv:2610.14549v1 Announce Type: cross Abstract: Physical systems and distributed AI workflows can retain past information, receive delayed feedback and combine conflicting signals. These features pose a common modeling problem: determining what information about past...

https://arxiv.org/abs/2610.14549

74 CONF

LLM-based User-Product Relevance Evaluation for Dynamic Product Advertising

ArXiv CS.AI 50%

arXiv:2610.14550v1 Announce Type: cross Abstract: Dynamic Product Advertising (DPA) retrieval systems are a core component of modern e-commerce advertising; however, their evaluation remains challenging. Conventional offline engagement metrics are limited by sparse, biased...

https://arxiv.org/abs/2610.14550

75 CONF

Prospective Forcing: Learning to Look Ahead for Autoregressive Long Video Diffusion

ArXiv CS.AI 50%

arXiv:2610.14654v1 Announce Type: cross Abstract: Autoregressive video generators synthesize long videos through chunk-wise rollouts, yet they are typically trained to predict only the immediate future. This limited prediction horizon makes it challenging to maintain video...

https://arxiv.org/abs/2610.14654

76 CONF

AlignAudio: A Unified Dual-Alignment Framework for Noise-Robust Speech-Conditioned Generation

ArXiv CS.AI 50%

arXiv:2610.14697v1 Announce Type: cross Abstract: Speech-conditioned generation maps speech signals to target audio outputs. Prior end-to-end speech-conditioned generative models achieve high fidelity in clean acoustic conditions, but their performance degrades substantially...

https://arxiv.org/abs/2610.14697

77 CONF

Forecasting Brain Tracer Distribution from Pre-Contrast MRI: A Comparison of Direct Regression and Flow Matching

ArXiv CS.AI 50%

arXiv:2609.22377v2 Announce Type: replace-cross Abstract: Intrathecal contrast-enhanced MRI shows how a tracer injected into the cerebrospinal fluid spreads through the brain, but requires repeated scans over 24--48\,h. We therefore ask how much of the later tracer distribution...

https://arxiv.org/abs/2609.22377

78 CONF

Self-improving data synthesis can scale embedding models across domains and tasks

ArXiv CS.AI 50%

arXiv:2610.14730v1 Announce Type: cross Abstract: Adapting embedding models to new tasks requires training data that reflect task-specific information needs and relevance criteria. However, manual data design is costly and difficult to scale. Here we introduce SYNTRA, a novel...

https://arxiv.org/abs/2610.14730

79 CONF

Training a Compact, Strong, and Realistic Text-to-Speech Model

ArXiv CS.AI 50%

arXiv:2610.14776v1 Announce Type: cross Abstract: Text-to-speech (TTS) systems are conventionally optimized for intelligibility and speaker similarity. However, neither objective captures the prosodic detail that distinguishes human performances from synthetic speech, leaving a...

https://arxiv.org/abs/2610.14776

80 CONF

Depth-progressive 3D seismic velocity model building via 2D generative diffusion models

ArXiv CS.AI 50%

arXiv:2610.14812v1 Announce Type: cross Abstract: Three-dimensional (3D) velocity model building (VMB) remains computationally challenging with conventional approaches and native 3D neural networks, largely due to the prohibitive memory and training cost of full 3D formulations....

https://arxiv.org/abs/2610.14812

81 CONF

Seamless Deployment of Machine Learning Inference at the Far Edge: An Empirical AWS IoT Greengrass Architecture for Industrial Predictive Maintenance

ArXiv CS.AI 50%

arXiv:2610.14815v1 Announce Type: cross Abstract: Industrial IoT sensors produce continuous, high-volume data streams that predictive maintenance needs to analyse in real time. Cloud-centric architectures send all raw telemetry to central servers, which costs latency, bandwidth...

https://arxiv.org/abs/2610.14815

82 CONF

AI-Native Product Management: A Vision

ArXiv CS.AI 50%

arXiv:2610.14823v1 Announce Type: cross Abstract: Artificial intelligence is increasingly used to support product management tasks. At the same time, AI-native research and development (R&D) may increase software realization capacity. Organizations may realize software faster...

https://arxiv.org/abs/2610.14823

83 CONF

Cross-Lingual Transfer of Evil Vectors Is Directionally Asymmetric

ArXiv CS.AI 50%

arXiv:2610.17880v1 Announce Type: cross Abstract: Persona vectors are linear directions in the residual-stream activations of LLMs that encode behavioral traits and enable fine-grained control over model behavior. Among them, the evil persona vector is particularly relevant to...

https://arxiv.org/abs/2610.17880

84 CONF

PrT-Flow: Probability-Tractable Flow Matching for Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2610.17667v1 Announce Type: cross Abstract: Diffusion models and Flow Matching can represent flexible distributions over robot actions, but many reinforcement learning algorithms also require the probability of each action. We introduce PrT-Flow (Probability-Tractable Flow...

https://arxiv.org/abs/2610.17667

85 CONF

Humans vs. Machines: A Comparison With AI Agents in Software Reverse Engineering

ArXiv CS.AI 50%

arXiv:2610.14960v1 Announce Type: cross Abstract: Can an autonomous AI agent reverse engineer a binary as well as a skilled human analyst? Do they adopt the same strategy, and can agents scale to real-world programs? We compare human and LLM-based software reverse engineering...

https://arxiv.org/abs/2610.14960

86 CONF

The Geometry of Language Model Logits: Generic Structure and Learned Departures

ArXiv CS.AI 50%

arXiv:2610.18520v1 Announce Type: cross Abstract: Autoregressive language models produce, at every prediction step, a vocabulary-sized logit vector whose structure determines the conditional next-token distribution. We empirically identify that the ordered vocabulary-head logits...

https://arxiv.org/abs/2610.18520

87 CONF

Mixing Regularization for Average-Reward Q-Learning and Q-Planning

ArXiv CS.AI 50%

arXiv:2610.14988v1 Announce Type: cross Abstract: The average-reward setting for reinforcement learning is challenging because its Bellman operator is known to contract in span only when the underlying Markov chain is mixing. Whereas existing work typically assumes a contraction...

https://arxiv.org/abs/2610.14988

88 CONF

UltraBERT: Beyond the Limit of BERT Pre-Training

ArXiv CS.AI 50%

arXiv:2610.15028v1 Announce Type: cross Abstract: Recent bidirectional encoders have been modernized with innovations from causal large language models (LLMs). However, the pretraining recipe itself remains inherited from small-scale conventions, leaving encoders underoptimized,...

https://arxiv.org/abs/2610.15028

89 CONF

Toward System One Clinical Intelligence: Parameter-Efficient Fine-Tuning and Counterfactual Explainability of Decision Foundation Models

ArXiv CS.AI 50%

arXiv:2610.15029v1 Announce Type: cross Abstract: Automated disease classification from patient-reported symptoms requires balancing the speed of lightweight discriminative models against the contextual understanding and latency costs of autoregressive large language models...

https://arxiv.org/abs/2610.15029

90 CONF

Select or Rewrite: Oracle-Guided Fine-Tuning for Generative ASR Post-Editing from N-Best Hypotheses

ArXiv CS.AI 50%

arXiv:2610.15056v1 Announce Type: cross Abstract: Large language models can post-edit ASR by using linguistic context to compare N-best hypotheses and rewrite the transcript when needed, often improving WER without retraining the recognizer. However, prompt-only correction is...

https://arxiv.org/abs/2610.15056

91 CONF

Deep Learning-based Semantic Segmentation of Dental Radiographs: Comparative Evaluation of PSPNet, DeepLabv3, UNet, and UNet++

ArXiv CS.AI 50%

arXiv:2610.15061v1 Announce Type: cross Abstract: Accurate and efficient analysis of dental radiographs has become increasingly important for disease diagnosis and clinical decision making. In particular, dental radiographs segmentation plays a critical role as it is challenging...

https://arxiv.org/abs/2610.15061

92 CONF

ShellElastic: Scalable Elastic Pretraining with Simultaneous Maximal Updates and Hyperparameter Transfer

ArXiv CS.AI 50%

arXiv:2610.15124v1 Announce Type: cross Abstract: Foundation models need to support diverse inference budgets, yet training a separate model for each budget duplicates pretraining effort. Elastic models train several nested sub-networks inside one weight-shared supernet, but...

https://arxiv.org/abs/2610.15124

93 CONF

To Defer or Not to Defer? Online Learning with LLMs and Congested Experts

ArXiv CS.AI 50%

arXiv:2610.15128v1 Announce Type: cross Abstract: Modern AI systems increasingly combine large language models (LLMs) with human experts, using the LLM to handle routine queries while deferring uncertain cases for expert review. When expert capacity is limited, however, deferral...

https://arxiv.org/abs/2610.15128

94 CONF

Framework for an AI-Ready ARPES Community Database

ArXiv CS.AI 50%

arXiv:2610.15236v1 Announce Type: cross Abstract: Condensed matter stands to gain profoundly from widespread adoption of artificial intelligence (AI). However, its current means of data organization impedes this endeavor. As a premier measure of electronic structure underlying...

https://arxiv.org/abs/2610.15236

95 CONF

North or South? Diagnosing LLMs' Understanding of North Korean Language, Society, and Knowledge

ArXiv CS.AI 50%

arXiv:2610.15282v1 Announce Type: cross Abstract: Large Language Models (LLMs) have made substantial progress in multilingual question answering, yet their ability to understand low-resource sociolinguistic contexts remains under-explored. We study North Korea as a unique...

https://arxiv.org/abs/2610.15282

96 CONF

Accuracy and Uncertainty in Machine Deception Detection (Research White Paper)

ArXiv CS.AI 50%

arXiv:2610.15314v1 Announce Type: cross Abstract: A deception detector is only as useful as the trust its alarms deserve. Like a polygraph, detectors that read a language model's internal activations can catch deceptive responses that look honest. Yet current detectors depend on...

https://arxiv.org/abs/2610.15314

97 CONF

Anchor-and-Harvest Curriculum: A Fully Unmasked Non-LLM Curriculum for Craftax

ArXiv CS.AI 50%

arXiv:2610.15328v1 Announce Type: cross Abstract: We introduce Anchor-and-Harvest Curriculum (AHC), a domain-informed, rule-based training system for Craftax-Full. AHC combines achievement-oriented tasks, harvested initial-state progress, difficulty adaptation, bounded...

https://arxiv.org/abs/2610.15328

98 CONF

ZOIF: Zeroth-Order Influence Function Estimation for LLM Data Selection Using Only Forward Passes

ArXiv CS.AI 50%

arXiv:2610.15336v1 Announce Type: cross Abstract: Influence functions enable principled data selection for large language models (e.g., identifying influential training samples for model editing, debugging, or efficient fine-tuning). However, scaling IFs to LLMs is hindered by...

https://arxiv.org/abs/2610.15336

99 CONF

CodeWalk: Generating Coding Benchmarks by Walking a Problem Graph

ArXiv CS.AI 50%

arXiv:2610.15345v1 Announce Type: cross Abstract: High-quality coding problems are expensive to collect. Benchmarks such as SWE-bench rely on substantial human curation, while synthetic alternatives typically rely on costly model generation and validation. In this work, we show...

https://arxiv.org/abs/2610.15345

00 CONF

In-Place Indic Tokenizer Expansion and Continual Pretraining of Nemotron-3-Nano-30B-A3B

ArXiv CS.AI 50%

arXiv:2610.15373v1 Announce Type: cross Abstract: We adapted Nemotron-3-Nano-30B-A3B for eight Indian languages without changing its backbone architecture. We expanded the vocabulary in place from 131,072 to 163,072 entries using continued byte-pair encoding and AdaptMix...

https://arxiv.org/abs/2610.15373

01 CONF

Security in LLM-Assisted Coding: A Systematic Literature Review

ArXiv CS.AI 50%

arXiv:2610.15482v1 Announce Type: cross Abstract: Large Language Models accelerate software development but introduce severe security vulnerabilities. This systematic review investigates security risks in automated coding, benchmark limitations, and the role of human developers....

https://arxiv.org/abs/2610.15482

02 CONF

REPRO: Reflective Evolutionary Prompt Optimizer for Automated LLM Red-Teaming

ArXiv CS.AI 50%

arXiv:2610.15513v1 Announce Type: cross Abstract: Safety-aligned large language models (LLMs) remain vulnerable to adversarial prompts that bypass their safety training. Existing automated red-teaming methods either require privileged access unavailable in production, rely on...

https://arxiv.org/abs/2610.15513

03 CONF

Dynamic Conflict Clustering with Lightweight Learning and Strategic Repair for Distributed Constraint Satisfaction

ArXiv CS.AI 50%

arXiv:2610.15658v1 Announce Type: cross Abstract: Distributed Constraint Satisfaction Problems (DCSPs) model distributed decision-making in multi-agent systems, where agents assign values to local variables while satisfying shared constraints. However, many DCSP solvers still...

https://arxiv.org/abs/2610.15658

04 CONF

Uncovering the Generative Landscape of Diffusion Models: A Spectral-Guided Quality-Diversity Diffusion Framework

ArXiv CS.AI 50%

arXiv:2610.15724v1 Announce Type: cross Abstract: Denoising Diffusion Probabilistic Models have emerged as powerful generative priors across domains ranging from molecular design to high-resolution image synthesis. However, standard diffusion sampling is inherently biased toward...

https://arxiv.org/abs/2610.15724

05 CONF

SkillMarker: A KGW-Based Watermarking Method for Agent Skill Ownership Verification

ArXiv CS.AI 50%

arXiv:2610.15725v2 Announce Type: cross Abstract: Agent skills encode operational procedures and domain expertise as reusable knowledge assets, yet their distribution in plaintext exposes them to unauthorized copying and leakage. However, practical mechanisms for skill ownership...

https://arxiv.org/abs/2610.15725

06 CONF

An Explainable System for Offensive Content Detection in Nastaliq Urdu: Corpus Curation, Transformer Benchmarking, and Word-Level Attribution

ArXiv CS.AI 50%

arXiv:2610.15795v1 Announce Type: cross Abstract: Automated moderation systems are deployed at scale, yet they remain opaque to the users and moderators affected by their decisions, and their coverage is almost entirely confined to English. For Urdu written in the Nastaliq...

https://arxiv.org/abs/2610.15795

07 CONF

Make Every Expansion Count: Learning to Search Efficiently in LLM Code Repair

ArXiv CS.AI 50%

arXiv:2610.15894v1 Announce Type: cross Abstract: Large language model (LLM) code-repair systems fix faulty programs by repeatedly generating and testing revisions. Each revision is costly, making the choice of which repair to pursue important for efficiency. Existing search...

https://arxiv.org/abs/2610.15894

08 CONF

What the Model Cannot See: Covert and Provable Text Invisibility in LLMs

ArXiv CS.AI 50%

arXiv:2610.15883v1 Announce Type: cross Abstract: Text wrapped in a trigger can be made behaviorally invisible to a language model: omitted from summaries and quotations, ignored in question answering, and left unflagged by downstream filtering. We study this selective blindness...

https://arxiv.org/abs/2610.15883

09 CONF

DiffNash: Finding Nash Equilibrium in High-Dimensional Continuous Action Spaces with Diffusion Policies

ArXiv CS.AI 50%

arXiv:2610.15932v1 Announce Type: cross Abstract: Computing equilibria in high-dimensional continuous games requires learning and representing rich, multi-modal strategy distributions, something conventional neural policies struggle to achieve. Diffusion models offer a natural...

https://arxiv.org/abs/2610.15932

10 CONF

Hardware-Assisted Multi-Stage Ransomware Detection with Adaptive Machine Learning

ArXiv CS.AI 50%

arXiv:2610.15951v1 Announce Type: cross Abstract: Ransomware is a fast-escalating privacy threat causing significant financial and data losses worldwide. As attacks grow more sophisticated, traditional static signatures and dynamic behavior analysis struggle to defend. In this...

https://arxiv.org/abs/2610.15951

11 CONF

Unfair and Unsafe: Everyday Bias Triggers Emergent Misalignment

ArXiv CS.AI 50%

arXiv:2610.15961v1 Announce Type: cross Abstract: Emergent misalignment (EM) occurs when fine-tuning a language model to a narrow task makes it misaligned on unrelated ones. Because organizations now routinely fine-tune models on their own data, it matters which kinds of...

https://arxiv.org/abs/2610.15961

12 CONF

Embedding Information in Policy Randomness: Distribution-Preserving Implicit Communication in MDPs

ArXiv CS.AI 50%

arXiv:2610.15976v1 Announce Type: cross Abstract: This paper investigates a new role of randomness in stochastic policies--as a medium for embedding and communicating information. Specifically, we consider an agent interacting with an environment according to a prescribed...

https://arxiv.org/abs/2610.15976

13 CONF

AutoSec-RAN: Specification-Grounded LLM Fine-Tuning for Automated Security Analysis of O-RAN

ArXiv CS.AI 50%

arXiv:2610.15995v1 Announce Type: cross Abstract: This paper presents AutoSec-RAN, a specification-grounded large language model (LLM) framework for automated security triage of Open Radio Access Network (O-RAN) testing. Unlike approaches relying on static rule sets or...

https://arxiv.org/abs/2610.15995

14 CONF

When Agents Talk, Secrets Travel: Topology-Driven Leakage and Recipient-Conditioned Sanitization in Multi-Agent LLM Systems

ArXiv CS.AI 50%

arXiv:2610.16039v1 Announce Type: cross Abstract: Multi-agent LLM systems (MALS) introduce a privacy-leakage problem that conventional sensitivity-based defenses miss, since information disclosed to one authorized agent can propagate through routine coordination to an...

https://arxiv.org/abs/2610.16039

15 CONF

Autoregressive Decoding as a Martingale: Characterizing the Stochastic Dynamics of LLM Generation

ArXiv CS.AI 50%

arXiv:2610.16041v1 Announce Type: cross Abstract: Large language models (LLMs) have achieved remarkable success across a wide range of language tasks. At their core, autoregressive LLMs are conditional probability models that learn to approximate the next-token distribution...

https://arxiv.org/abs/2610.16041

16 CONF

AI Representations of Mental Health Stigma: Quantifying the Gap Between AI Estimations and Human Perceptions Across Gender Groups Using GPT-Based LLMs

ArXiv CS.AI 50%

arXiv:2610.16042v1 Announce Type: cross Abstract: The increasing use of artificial intelligence (AI) in mental health applications raises concerns about its potential to internalize and perpetuate societal biases. This research evaluates both disparities and misalignment in how...

https://arxiv.org/abs/2610.16042

17 CONF

When Agent Evaluation Becomes Unreliable: An Empirical Study of Oracle Failures and Configuration-Dependent Measurement in an LLM Agent Benchmark

ArXiv CS.AI 50%

arXiv:2610.16067v1 Announce Type: cross Abstract: Agent evaluation harnesses are software systems that run test suites, generate inputs, and decide outcomes with test oracles. This paper audits such software for reliability defects that score tables cannot reveal. We introduce...

https://arxiv.org/abs/2610.16067

18 CONF

Where Polanyi Meets Galois: Tacit Knowing as an Algebro-Deterministic Operation, Toward Re-rooting AI-Generated Science

ArXiv CS.AI 50%

arXiv:2610.16102v1 Announce Type: cross Abstract: We argue that tacit knowing is better modelled as an operating regime of one memory substrate than as a kind of content. The substrate is VaCoAl, an algebro-deterministic XOR-and-shift hyperdimensional memory over GF(2), realized...

https://arxiv.org/abs/2610.16102

19 CONF

OracleProto: A Reproducible Framework for Benchmarking LLM Forecasting via Knowledge Cutoffs and Temporal Masking

ArXiv CS.AI 50%

arXiv:2605.03762v2 Announce Type: replace Abstract: Forecasting is a composite capability encompassing information gathering, evidence integration, situational assessment, and decision-making. Large language models (LLMs) combine reasoning and tool use, offering new avenues for...

https://arxiv.org/abs/2605.03762

20 CONF

Omni-Modal LLMs Know More Than They Say

ArXiv CS.AI 50%

arXiv:2610.16135v1 Announce Type: cross Abstract: Omni-modal large language models (Omni-LLMs) extend LLMs to process heterogeneous modalities (e.g., vision, audio, and IMU signals) alongside language, enabling a broader range of multimodal tasks. Yet we find that these models...

https://arxiv.org/abs/2610.16135

21 CONF

Fisher information and the geometry of memorization in neural networks

ArXiv CS.AI 50%

arXiv:2610.16161v1 Announce Type: cross Abstract: A fruitful approach towards understanding generalization in machine learning involves characterizing how models separate generalizable structure from idiosyncratic or even corrupted features of training data. Fisher information...

https://arxiv.org/abs/2610.16161

22 CONF

MP-MNCA: Morphology-Prior Masked Neighbor Cross-Attention for Efficient Spatial Domain Identification

ArXiv CS.AI 50%

arXiv:2610.16212v1 Announce Type: cross Abstract: Spatial transcriptomics jointly captures molecular state, tissue morphology, and spatial organization, yet spatial-domain methods are typically compared only at late training checkpoints, obscuring differences in convergence...

https://arxiv.org/abs/2610.16212

23 CONF

WorldRoPE: Probabilistic World-Centric Rotary Embedding for Multi-View Attention

ArXiv CS.AI 50%

arXiv:2610.16337v1 Announce Type: cross Abstract: Rotary Positional Embedding (RoPE) is widely used to encode relative positions in Transformers. In multi-view settings, however, patch coordinates are defined independently in each image: the same 3D point can therefore have...

https://arxiv.org/abs/2610.16337

24 CONF

From Caricature to Collective Artifact: How Persona Construction Drives Polarization in Generative Social Simulations

ArXiv CS.AI 50%

arXiv:2610.16498v1 Announce Type: cross Abstract: Generative Social Simulation (GSS) leverages Large Language Models (LLMs) to create sophisticated personas that define agents with demographics, emotions, and political opinions. While prior work shows that personas compress...

https://arxiv.org/abs/2610.16498

25 CONF

Towards Modality-Agnostic Seizure Classification: A Unified Deep Learning Framework for Heterogeneous EEG and Intracranial EEG Signals

ArXiv CS.AI 50%

arXiv:2610.16596v2 Announce Type: cross Abstract: Seizure prediction and detection, essential for clinical treatment of epilepsy, remain the crucial task of this neurological disease. EEG and iEEG are widely used to analyze epilepsy due to its capability of capturing...

https://arxiv.org/abs/2610.16596

26 CONF

Learning Incompressible 3D Flows from Sparse Samples: A Set Transformer Approach to Divergence-Free Kernel Fitting

ArXiv CS.AI 50%

arXiv:2610.16645v1 Announce Type: cross Abstract: Recovering divergence-free velocity fields from sparse and corrupted observations requires reconciling physical consistency, expressive representations, and robustness to missing data. We introduce DFK-Former, a set-to-field...

https://arxiv.org/abs/2610.16645

27 CONF

StableJudge: A Lightweight Adversarial Defense Framework for LLM-as-a-Judge

ArXiv CS.AI 50%

arXiv:2610.16814v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly used as judges to assign scalar scores that drive downstream decision-making, including tasks such as paper review and recommendation. However, LLM-as-a-Judge pipelines are vulnerable...

https://arxiv.org/abs/2610.16814

28 CONF

DTA-GT: Direction- and Topology-Aware Graph Transformer for Neural Network Representation Learning

ArXiv CS.AI 50%

arXiv:2610.16822v1 Announce Type: cross Abstract: Architecture attribute predictors reduce the cost of neural architecture search by estimating accuracy or latency from a small set of evaluated candidates. The key difficulty is that neural architectures are operation-labeled...

https://arxiv.org/abs/2610.16822

29 CONF

KNN Implementation Details Can Dramatically Change Performance: An Example from Cover Trees

ArXiv CS.AI 50%

arXiv:2610.16940v1 Announce Type: cross Abstract: Nearest neighbor search algorithms are widely used and have a long history, but we find that key implementation details are seldom reviewed in articles about them. These details can materially impact performance, and thus change...

https://arxiv.org/abs/2610.16940

30 CONF

Proactive Physical AI through Continuous Observation and Timely Intervention

ArXiv CS.AI 50%

arXiv:2610.16967v1 Announce Type: cross Abstract: Physical AI is commonly formulated around an explicitly specified task: given a goal or instruction, a system perceives, reasons, and acts. In many real-world settings, however, important tasks emerge from continuously changing...

https://arxiv.org/abs/2610.16967

31 CONF

Generative Flow on Symbolic Abstractions for Embodiment Capability Discovery

ArXiv CS.AI 50%

arXiv:2610.17174v1 Announce Type: cross Abstract: Achieving complex, long-horizon robot behavior remains challenging because successful execution depends on skills grounded in a robot's physical affordances. Existing training-task distributions are often handcrafted or sampled...

https://arxiv.org/abs/2610.17174

32 CONF

What the Machine Invented: Mining New Mathematical Instruments from OpenAI's October 2026 Release of AI-Generated Mathematics

ArXiv CS.AI 50%

arXiv:2610.17090v1 Announce Type: cross Abstract: On 6 October 2026 OpenAI published an initial 722 AI-written mathematical manuscripts, organized into 372 problem families, in the public repository github.com/openai/math. The release includes claimed solutions to many prominent...

https://arxiv.org/abs/2610.17090

33 CONF

AIPOCH Open-Science: An Open-Source AI Research Workbench with Reproducible .science Packages

ArXiv CS.AI 50%

arXiv:2610.17110v1 Announce Type: cross Abstract: AI-assisted research leaves instructions, computations, artifact revisions and checks whose relationships matter when another researcher receives the work. We present AIPOCH Open-Science, an open-source research workbench with...

https://arxiv.org/abs/2610.17110

34 CONF

Who is the 'I' in AI? Establishing Decentralized Identity and Accountability for Autonomous Agents

ArXiv CS.AI 50%

arXiv:2610.17126v1 Announce Type: cross Abstract: The rapid proliferation of autonomous AI agents in decentralized ecosystems challenges existing identity paradigms. While human identity is traditionally anchored in biological and/or self-asserted attributes and government...

https://arxiv.org/abs/2610.17126

35 CONF

Deleting Code Without Losing Knowledge: Regenerative Software for Python Libraries

ArXiv CS.AI 50%

arXiv:2610.17197v1 Announce Type: cross Abstract: In the age of "Software 3.0", when a complete program can be rewritten in minutes by an agent, the traditional approach of relying on code as the sole source of truth becomes questionable. Fowler advocated for a set of seven...

https://arxiv.org/abs/2610.17197

36 CONF

Thinking Outside the Bots: Human Ideas in AI's Semantic Blind Spots Are Not Inferior

ArXiv CS.AI 50%

arXiv:2610.17324v1 Announce Type: cross Abstract: AI-generated outputs are frequently perceived as highly creative, but also tend to be more similar to each other (homogenized) than human ideas. Does the human heterogeneity that extends beyond AI reflect valuable thinking that...

https://arxiv.org/abs/2610.17324

37 CONF

Spec-Harness: Assessing and Guiding LLM-Synthesized Formal Specifications Beyond Verifiability

ArXiv CS.AI 50%

arXiv:2604.00280v3 Announce Type: replace-cross Abstract: Formal specifications play a central role in ensuring software reliability, yet automatically synthesizing high-quality specifications remains difficult and often requires domain expertise. Recent work has applied large...

https://arxiv.org/abs/2604.00280

38 CONF

Event-Centric Earthquake Monitoring with a Multi-Station Transformer

ArXiv CS.AI 50%

arXiv:2610.17404v1 Announce Type: cross Abstract: Earthquake catalogs underpin studies such as fault structure, earthquake processes and seismic hazard, and their value depends on being both complete and reliable. Machine learning has improved individual tasks of earthquake...

https://arxiv.org/abs/2610.17404

39 CONF

Why Agentic Workflows Fail: A Taxonomy Derived from Literature and Empirical Data

ArXiv CS.AI 50%

arXiv:2610.17418v1 Announce Type: cross Abstract: Agentic workflows combine large language models with planning, memory, tools, execution environments, and iterative feedback. Their failures are difficult to diagnose because the same user-visible symptom may originate from model...

https://arxiv.org/abs/2610.17418

40 CONF

Assessing the Performance of AI Models in Misinformation Detection Using Predefined Social Noise Constructs

ArXiv CS.AI 50%

arXiv:2610.17469v1 Announce Type: cross Abstract: Social media platforms generate vast volumes of user-generated content that often obscure meaningful discourse, a phenomenon conceptualized as social noise. While existing misinformation detection methods primarily focus on...

https://arxiv.org/abs/2610.17469

41 CONF

A Few Steps Further: Why Defenses Against Malicious Finetuning Erode Under Continued Training

ArXiv CS.AI 50%

arXiv:2605.14605v4 Announce Type: replace-cross Abstract: Model providers increasingly release the weights of large language models. Although these models are safety-aligned before release, their safeguards can often be removed by fine-tuning on harmful data. A growing class of...

https://arxiv.org/abs/2605.14605

42 CONF

Towards Green AI: Evaluating Energy Efficiency for Quantized LLM Inference

ArXiv CS.AI 50%

arXiv:2610.17531v1 Announce Type: cross Abstract: While it is known that quantization of large language models (LLMs) reduces memory usage via lower-bit weights, studies quantifying the resulting impact on energy usage and carbon intensity are scarce. We present the first...

https://arxiv.org/abs/2610.17531

43 CONF

Dissociating Answerability from Extraction in LLM Question Answering over Radiology Reports

ArXiv CS.AI 50%

arXiv:2610.17576v1 Announce Type: cross Abstract: Answering questions from radiology reports involves two decisions: whether the report contains an answer and which span gives it. A false answer presents an unsupported claim as fact; an over-abstention hides what the report...

https://arxiv.org/abs/2610.17576

44 CONF

Tanawpamana: Benchmarking Vision-Language Models on Philippine Cultural Heritage Sites

ArXiv CS.AI 50%

arXiv:2610.17596v1 Announce Type: cross Abstract: Vision-language models are increasingly used to identify and describe places, yet their recognition and knowledge of specific cultures remain poorly understood, particularly in underrepresented regions such as the Philippines....

https://arxiv.org/abs/2610.17596

45 CONF

Morphology-aware deep learning predicts colorectal cancer survival from whole-slide images

ArXiv CS.AI 50%

arXiv:2610.17603v1 Announce Type: cross Abstract: Background: Five-year survival prediction from hematoxylin-and-eosin whole-slide images could support risk stratification, but the task is weakly supervised, and the tissue morphologies carrying prognostic information remain...

https://arxiv.org/abs/2610.17603

46 CONF

The Tokens Strike Back: Safety-Aware Denoising with Directional Embedding-based Revision

ArXiv CS.AI 50%

arXiv:2610.17636v1 Announce Type: cross Abstract: The Safety-Aware Denoiser (SAD) is a recent, training-free, safety-guidance framework for text diffusion models. During inference, SAD steers text samples toward safe regions of the token probability space. In this paper, we...

https://arxiv.org/abs/2610.17636

47 CONF

HelpMeOut: A Neuropsychology-Grounded Benchmark for Error-Aware Monitoring of Household Task Execution

ArXiv CS.AI 50%

arXiv:2610.17694v1 Announce Type: cross Abstract: We introduce HelpMeOut, an embodied AI benchmark for error-aware monitoring of household task execution. Existing embodied household benchmarks typically assume that humans execute tasks correctly, but this assumption does not...

https://arxiv.org/abs/2610.17694

48 CONF

Explore with Memory, Learn without It: In-Context Exploration and Self-Distillation for LLM Reasoning

ArXiv CS.AI 50%

arXiv:2610.17734v1 Announce Type: cross Abstract: Exploration bounds learning efficiency in reinforcement learning with verifiable rewards (RLVR). Current methods explore mostly through independent sampling, without fully using the information gathered along the way, such as...

https://arxiv.org/abs/2610.17734

49 CONF

Train for Many, Update with One: Improving Multi-Sample Reasoning via Fenchel-Bregman Posterior Learning

ArXiv CS.AI 50%

arXiv:2610.17735v1 Announce Type: cross Abstract: Multi-sample reasoning measures candidate coverage, the probability that at least one of K candidates is correct. However, minimizing the average loss of individual attempts does not directly optimize candidate coverage. We...

https://arxiv.org/abs/2610.17735

50 CONF

Gaussian Process Bandits for AI System Optimization with Selectively Verified Feedback

ArXiv CS.AI 50%

arXiv:2610.17760v1 Announce Type: cross Abstract: Gaussian process bandits exploit similarities among configurations to optimize prompts, retrieval systems, and agent workflows. In these applications, however, inexpensive model-based evaluations can disagree with trusted task...

https://arxiv.org/abs/2610.17760

51 CONF

Same Attack, Different Judge: A First Controlled Adversarial Comparison of Jev, LLM Judges, and Purpose-Built Safety Classifiers

ArXiv CS.AI 50%

arXiv:2610.17820v1 Announce Type: cross Abstract: TypeSafe AI's Jev is a new kind of safety judge. It generates no text and no label: it answers questions the developer defines in advance and returns a probability for each answer. The vendor admits that adversarial input can...

https://arxiv.org/abs/2610.17820

52 CONF

A Tutorial on Quantum Amplitude Embedding with Data Re-Uploading

ArXiv CS.AI 50%

arXiv:2610.17836v1 Announce Type: cross Abstract: Amplitude embedding lets a variational quantum model read many classical features with few qubits, but the model output is then only a quadratic form in the normalized input. Data re-uploading, which encodes the input repeatedly...

https://arxiv.org/abs/2610.17836

53 CONF

ProtoArena: Can AI Agents Discover and Exploit Real-World Cross-Component Protocol Vulnerabilities?

ArXiv CS.AI 50%

arXiv:2610.17837v1 Announce Type: cross Abstract: Large language model (LLM) agents are increasingly used for vulnerability analysis, yet existing benchmarks provide limited coverage of exploitation that depends on interactions among network protocol components. Such...

https://arxiv.org/abs/2610.17837

54 CONF

Uni-RS: A Spatially Faithful Unified Understanding and Generation Model for Remote Sensing

ArXiv CS.AI 50%

arXiv:2601.17673v2 Announce Type: replace-cross Abstract: Unified remote sensing multimodal models exhibit a pronounced spatial reversal curse: Although they can accurately recognize and describe object locations in images, they often fail to faithfully execute the same spatial...

https://arxiv.org/abs/2601.17673

55 CONF

GLST: Global-Local Spatio-Temporal Memory Decomposition Framework for Online Chained-Goal Navigation

ArXiv CS.AI 50%

arXiv:2610.17927v1 Announce Type: cross Abstract: Autonomous navigation in real-world environments often requires long-horizon operation under partial observability and evolving goals. Existing mapless navigation approaches predominantly assume fixed goals throughout an episode,...

https://arxiv.org/abs/2610.17927

56 CONF

AI-Enhanced Hyperspectral Imaging for Intraoperative Detection of Low-Grade Glioma: A Clinical Study

ArXiv CS.AI 50%

arXiv:2610.17944v1 Announce Type: cross Abstract: Current image-guided neuronavigation techniques, such as intraoperative MRI and ultrasound, are limited by high costs, limited sensitivity and operational complexity. There is a clear need for more efficient systems to enhance...

https://arxiv.org/abs/2610.17944

57 CONF

Better Capturing Interactions Between Products in~Retail: Revisited Negative Sampling for~Basket Choice Modeling

ArXiv CS.AI 50%

arXiv:2610.17963v1 Announce Type: cross Abstract: Brick-and-mortar retailers face many different challenges that involve understanding thoroughly its products catalog and customer preferences. In particular, assortment optimization - proposing the ideal mix of products - and...

https://arxiv.org/abs/2610.17963

58 CONF

Seeing Through the Forgery Is Not Enough: Regressive Sycophancy under Fabricated Evidence in Medical Vision-Language Models

ArXiv CS.AI 50%

arXiv:2610.17989v1 Announce Type: cross Abstract: A medical vision-language model may analyze a given CT scan and correctly detect both kidneys in view. Continuing this conversation, if a user provides a prescription with an opposite direction, medical VLMs often sycophantically...

https://arxiv.org/abs/2610.17989

59 CONF

Cost-Efficient Privacy Enforcement for Email-Generating LLM Agents under Strategic Information Requests

ArXiv CS.AI 50%

arXiv:2610.18007v1 Announce Type: cross Abstract: LLM agents now act as delegated communication intermediaries: they read private context, draft messages, and send externally visible emails on behalf of a user. Existing checker-repair defenses can reduce inappropriate...

https://arxiv.org/abs/2610.18007

60 CONF

Scaling Vision Transformers Without Training: Prune-and-Merge Token Compression

ArXiv CS.AI 50%

arXiv:2610.18039v2 Announce Type: cross Abstract: Vision Transformers achieve strong performance but incur high computational costs from quadratic token interactions. Training-free token reduction methods improve efficiency, but their robustness remains underexplored. We propose...

https://arxiv.org/abs/2610.18039

61 CONF

H-DDCM: Hadamard Codebooks for Fast Generative Image and Video Compression

ArXiv CS.AI 50%

arXiv:2610.18125v1 Announce Type: cross Abstract: Training-free diffusion-based compression methods repurpose the generative priors of a diffusion or a flow model to compress data without retraining them as task-specific codecs. Recent codebook-based approaches such as...

https://arxiv.org/abs/2610.18125

62 CONF

Claim Framing as a Driver of LLM Sycophancy

ArXiv CS.AI 50%

arXiv:2610.18159v1 Announce Type: cross Abstract: Sycophancy in Large Language Models, the tendency to align responses with user preferences over ground truth, has largely been studied using questions without a clear ground truth and with a single, plainly stated user opinion....

https://arxiv.org/abs/2610.18159

63 CONF

Lifting Finite-Sample Gaussian Approximation of Wide Neural Networks to Functional Convergence

ArXiv CS.AI 50%

arXiv:2610.18166v1 Announce Type: cross Abstract: Randomly initialised neural networks of width m converge to Gaussian processes as $m \to \infty$, with two types of quantitative bounds characterising this convergence. Finite-sample bounds compare the network with its Gaussian...

https://arxiv.org/abs/2610.18166

64 CONF

A Controlled Evaluation of Curvature Inspired Scalar Feature Scaling for Domain Generalization

ArXiv CS.AI 50%

arXiv:2610.18227v1 Announce Type: cross Abstract: Domain-specific curvature is an appealing idea for domain generalization (DG), but curvature learning can denote substantially different computations. This paper evaluates one deliberately narrow implementation: the code calls it...

https://arxiv.org/abs/2610.18227

65 CONF

Heading Hierarchy Reconstruction for Complex and Long Documents with Level-Wise Multimodal Memory

ArXiv CS.AI 50%

arXiv:2610.18238v1 Announce Type: cross Abstract: Vision-language models (VLMs) have been widely adopted for document parsing and optical character recognition (OCR), as they can accurately extract page-level elements. However, downstream applications such as retrieval-augmented...

https://arxiv.org/abs/2610.18238

66 CONF

LiFUSE: Frequency-Spatial Unified Self-Supervised Pretraining for Liver Disease Analysis on Non-contrast CT

ArXiv CS.AI 50%

arXiv:2610.18243v1 Announce Type: cross Abstract: Given the growing global burden of liver tumors and diffuse liver diseases, non-contrast computed tomography (NCCT) is increasingly used for opportunistic screening because of its low cost and favorable safety profile. However,...

https://arxiv.org/abs/2610.18243

67 CONF

FALLNEST: Dual-Domain Dataset Condensation of Imbalanced IMU Data for Wearable Fall Detection

ArXiv CS.AI 50%

arXiv:2610.18260v1 Announce Type: cross Abstract: Wearable fall detection is hampered by rare fall events, scarce labeled IMU data, and the cost of training deep models on large, imbalanced corpora. Existing methods either rely on hand-crafted features with shallow classifiers,...

https://arxiv.org/abs/2610.18260

68 CONF

Encrypted Skills Can Become New Security Exploits If AI Agents Know How to Decipher

ArXiv CS.AI 50%

arXiv:2610.18262v1 Announce Type: cross Abstract: Third-party skills can expose LLM agents to malicious instructions hidden by encoding. We introduce Ciphered Skill Injection (CSI), which embeds such instructions in skill files for agents to decode and follow. We evaluate Caesar...

https://arxiv.org/abs/2610.18262

69 CONF

JevGPT: Thinking Fast and Slow in Autonomous Research

ArXiv CS.AI 50%

arXiv:2610.18272v1 Announce Type: cross Abstract: Autonomous research agents must acquire and reason over evidence within a limited budget. Yet relevant sources can remain uninspected or unused, leaving a gap between retrieval coverage and the evidence that informs research. The...

https://arxiv.org/abs/2610.18272

70 CONF

Which Optimizer Leaks Most? Optimizer Implicit Bias as a Training-Data Privacy Controller

ArXiv CS.AI 50%

arXiv:2610.18294v1 Announce Type: cross Abstract: Releasing a neural network checkpoint can leak information about its training data, and existing reconstruction results rely on gradient-descent training. We ask whether the choice of optimizer affects the ability to recover the...

https://arxiv.org/abs/2610.18294

71 CONF

Lightweight but Linguistically Fair? Quantisation Robustness across Bavarian, German, and English

ArXiv CS.AI 50%

arXiv:2610.18343v1 Announce Type: cross Abstract: Post-training quantisation reduces the memory cost of large language models, but it remains unclear whether compression preserves language skills of regional varieties as reliably as their standard-language counterparts. We study...

https://arxiv.org/abs/2610.18343

72 CONF

Do Factuality Methods Make LLMs More Factual? Reproducibility and Fair Comparison

ArXiv CS.AI 50%

arXiv:2610.18415v1 Announce Type: cross Abstract: Large Language Models (LLMs) are aligned to human preferences, after pre-training and supervised fine-tuning steps. However, standard RL alignment steps such as RLHF do not focus on factuality of LLMs, hence they may generate...

https://arxiv.org/abs/2610.18415

73 CONF

Agaram 2.5B: The First Tamil-First Language Model Trained from Scratch with Tool Calling. A Playbook for Under-Represented Languages

ArXiv CS.AI 50%

arXiv:2610.18420v1 Announce Type: cross Abstract: We trained Agaram 2.5B, a 2.5-billion-parameter Tamil-English language model, from scratch on about 50 billion tokens (58% Tamil, the rest English, math, code and parallel text) on one node of eight H100 GPUs in about six days,...

https://arxiv.org/abs/2610.18420

74 CONF

FedGraph: Defending Federated Large Language Model Fine-Tuning Against Backdoor Attacks via Graph-Based Aggregation

ArXiv CS.AI 50%

arXiv:2610.18430v1 Announce Type: cross Abstract: Federated fine-tuning of large language models (LLMs) allows multiple clients to collaboratively adapt models without sharing raw data, offering strong privacy benefits. However, this setting is highly vulnerable to backdoor...

https://arxiv.org/abs/2610.18430

75 CONF

How Well Do LLMs Perform in Peer Review? Evidence from Finance and Accounting

ArXiv CS.AI 50%

arXiv:2610.18435v1 Announce Type: cross Abstract: Referees increasingly use large language models (LLMs), while finance and accounting journals disagree about whether to allow it. We ask which part of a referee's job the models can do. GPT-4o, whose training data end before the...

https://arxiv.org/abs/2610.18435

76 CONF

A Global View of How LLMs Value Human Time

ArXiv CS.AI 50%

arXiv:2610.18515v2 Announce Type: cross Abstract: Large language models (LLMs) increasingly act as personal assistants; however, little is known about how they capture national-level preferences on a global scale. Given that "time" is a universal and representative concept, we...

https://arxiv.org/abs/2610.18515

77 CONF

The Many Morals of AI: The cautionary bias of North American news narratives in global perspective

ArXiv CS.AI 50%

arXiv:2610.18573v1 Announce Type: cross Abstract: The stories told about artificial intelligence (AI) play a central role in shaping public understanding, policy debates, and expectations about technological change. News media are especially influential in this process. Using a...

https://arxiv.org/abs/2610.18573

78 CONF

Geometric Calibration of Supervised Fine-Tuning

ArXiv CS.AI 50%

arXiv:2610.18634v1 Announce Type: cross Abstract: Recent studies investigate the weight geometry through its spectral structure, which characterizes how the weights transform different input directions. Within this view, supervised fine-tuning (SFT) is shown to induce dense...

https://arxiv.org/abs/2610.18634

79 CONF

Kwai Summary Attention V2 Technical Report

ArXiv CS.AI 50%

arXiv:2610.18654v1 Announce Type: cross Abstract: Summary-token architectures such as Kwai Summary Attention (KSA) compress distant context into learnable Summary token per chunk, and hybrid variants interleave these layers with full attention to retain exact access to the...

https://arxiv.org/abs/2610.18654

80 CONF

Model-Based Online Decision Making via Generative Trajectory Planning

ArXiv CS.AI 50%

arXiv:2610.18677v1 Announce Type: cross Abstract: Online decision-making faces a tension between efficient inference and long-horizon reasoning. Value-based methods rely on short-horizon bootstrapping that limits the propagation of long-term information, while trajectory...

https://arxiv.org/abs/2610.18677

81 CONF

Just Respect the Semantics: Image-Token Compression via Pre-LLM Semantic Variable-Resolution Tokenization and In-LLM Relevance Pruning

ArXiv CS.AI 50%

arXiv:2610.18690v1 Announce Type: cross Abstract: Image-token compression must reduce computation without hurting the semantics. We introduce Varest, guided by semantic fungibility: tokens within a coherent semantic region can be summarized coarsely, while object boundaries and...

https://arxiv.org/abs/2610.18690

82 CONF

SoFT: Soft Targets for Generalizable LLM Fine-Tuning

ArXiv CS.AI 50%

arXiv:2609.32493v2 Announce Type: replace-cross Abstract: Distillation enables student language models to acquire new capabilities from expert teachers. However, integrating knowledge from multi-teacher, multi-domain demonstrations into a single student remains challenging. We...

https://arxiv.org/abs/2609.32493

83 CONF

Where Do LLM Ensemble Gains Come From? Decomposing Candidate Value, Diversity, and Aggregation

ArXiv CS.AI 50%

arXiv:2610.18786v1 Announce Type: cross Abstract: LLM ensemble gains are often measured against a single baseline, but the value attributed to additional candidates can change with the comparison used. We study an LLM aggregator under four matched conditions: Direct (no...

https://arxiv.org/abs/2610.18786

84 CONF

Bytes to Threats: Byte-Sequence Transformer For Malware Detection

ArXiv CS.AI 50%

arXiv:2610.18787v1 Announce Type: cross Abstract: Existing malware detection approaches heavily rely on handcrafted features derived from static or dynamic analysis of the executable files. These methods require specialized domain expertise, dedicated analysis tools, and...

https://arxiv.org/abs/2610.18787

85 CONF

Back-Translation Still Matters: Leveraging Monolingual Data for Low-Resource Machine Translation with LLMs

ArXiv CS.AI 50%

arXiv:2610.18818v2 Announce Type: cross Abstract: Pre-trained Large Language Model (LLMs) are effective on many tasks in zero-shot and few-shot settings. For machine translation of low-resource languages, the obvious options are few-shot prompting and fine-tuning on a small...

https://arxiv.org/abs/2610.18818

86 CONF

Two Clocks: Why Larger Rollout Batches Learn Slower but Fail on Schedule in Reinforcement Learning with Verifiable Rewards

ArXiv CS.AI 50%

arXiv:2610.18853v1 Announce Type: cross Abstract: Reinforcement learning with verifiable rewards (RLVR) relies on costly model-generated responses, motivating larger rollout batches and square-root learning-rate scaling. Reported to approximately preserve progress per sample...

https://arxiv.org/abs/2610.18853

87 CONF

What Matters in Quality-Aware VLA Pretraining: A Benchmark and Empirical Study

ArXiv CS.AI 50%

arXiv:2610.18869v1 Announce Type: cross Abstract: How should demonstration quality be used during vision-language-action (VLA) pretraining to improve performance after target-task fine-tuning? We compare weighting, filtering, and conditioning across reward definitions and label...

https://arxiv.org/abs/2610.18869

88 CONF

Decidable Reasoning About Time in Finite-Domain Situation Calculus Theories

ArXiv CS.AI 50%

arXiv:2402.03164v3 Announce Type: replace Abstract: Representing time is crucial for cyber-physical systems and has been studied extensively in the situation calculus. The most commonly used approach represents time by adding a real-valued function $\mathit{time}(a)$ that...

https://arxiv.org/abs/2402.03164

89 CONF

A Governed Agent Architecture for Natural-Language Dashboard Generation over Enterprise DataBrain

ArXiv CS.AI 50%

arXiv:2610.06914v2 Announce Type: replace Abstract: We present a DataBrain-specific prototype that turns natural-language analytic requests into inspectable dashboards. An installable Codex plugin and standalone Agent Runtime combine schema-constrained model decisions with typed...

https://arxiv.org/abs/2610.06914

90 CONF

OPTIC-ER: A Reinforcement Learning Framework for Real-Time Emergency Response and Equitable Resource Allocation in Underserved African Communities

ArXiv CS.AI 50%

arXiv:2508.12943v3 Announce Type: replace Abstract: Public service systems in many African regions suffer from delayed emergency response and spatial inequity, causing avoidable suffering. This paper introduces OPTIC-ER, a reinforcement learning (RL) framework for real-time,...

https://arxiv.org/abs/2508.12943

91 CONF

Alignment Faking - the Train -> Deploy Asymmetry: Through a Game-Theoretic Lens with Bayesian-Stackelberg Equilibria

ArXiv CS.AI 50%

arXiv:2511.17937v2 Announce Type: replace Abstract: Alignment faking is a form of strategic deception in AI in which models selectively comply with training objectives when they infer that they are in training, while preserving different behavior outside training. The phenomenon...

https://arxiv.org/abs/2511.17937

92 CONF

GDEPO: Group Dual-dynamic and Equal-right Advantage Policy Optimization for Sample-Efficient Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2601.06795v4 Announce Type: replace Abstract: Automated Theorem Proving (ATP) is a fundamental AI challenge, requiring machine-verifiable proofs in formal languages like Lean to evaluate reasoning. Reinforcement Learning (RL), especially Group Relative Policy Optimization...

https://arxiv.org/abs/2601.06795

93 CONF

Stochastic CHAOS: Why Deterministic Inference Kills, and Distributional Variability Is the Heartbeat of Artifical Cognition

ArXiv CS.AI 50%

arXiv:2601.07239v2 Announce Type: replace Abstract: Deterministic inference is a comforting ideal in classical software: the same program on the same input should always produce the same output. As large language models move into real-world deployment, this ideal has been...

https://arxiv.org/abs/2601.07239

94 CONF

Strategy Executability in Mathematical Reasoning: Leveraging Human-Model Differences for Effective Guidance

ArXiv CS.AI 50%

arXiv:2602.22583v2 Announce Type: replace Abstract: Example-based guidance is widely used to improve mathematical reasoning at inference time, yet its effectiveness is highly unstable across problems and models-even when the guidance is correct and problem-relevant. We show that...

https://arxiv.org/abs/2602.22583

95 CONF

Argumentation-Based Person-Tailored Storytelling for Older Adults: A Reflective Hybrid AI Architecture

ArXiv CS.AI 50%

arXiv:2605.10531v2 Announce Type: replace Abstract: This work investigates knowledge-driven, large language model (LLM)-based storytelling for older adults. We present a reflective storytelling agent integrating knowledge graphs, user modelling, and argumentation theory to guide...

https://arxiv.org/abs/2605.10531

96 CONF

Energy per Successful Goal: Goal-Level Energy Accounting for Agentic AI Systems

ArXiv CS.AI 50%

arXiv:2605.22883v2 Announce Type: replace Abstract: Current AI energy benchmarks measure consumption per model invocation. For agentic systems, where a single user goal may trigger multistep orchestration, tool calls, retries, and failure recovery, the invocation count is an...

https://arxiv.org/abs/2605.22883

97 CONF

What a Pooled Leaderboard Does Not Say: Auditing the Scope and Provenance of Method Comparisons in Microservice RCA Benchmarks

ArXiv CS.AI 50%

arXiv:2606.29159v2 Announce Type: replace Abstract: Offline benchmarks for microservice root cause analysis (RCA) rank methods by a score pooled over several applications, and a pooled lead leaves two things unstated: on which of the systems it holds, and how the score was...

https://arxiv.org/abs/2606.29159

98 CONF

Agents Don't Just Agree, They Remember: Benchmarking Persistent Sycophancy in Self-Improving Personal Agents

ArXiv CS.AI 50%

arXiv:2607.10526v5 Announce Type: replace Abstract: Self-improving personal agents write profiles, memories, and reusable skills that carry over from one chat to the next. Prior works ask whether a model agrees with the user in its next reply. Yet these agents can also write...

https://arxiv.org/abs/2607.10526

99 CONF

MIND: Lightweight and Effective Memory Injection Defense for LLM Agents via Intent-Aware Information Bottleneck

ArXiv CS.AI 50%

arXiv:2607.28103v2 Announce Type: replace Abstract: Memory-augmented LLM-based agents are vulnerable to memory injection attacks: Agents may retrieve poisoned memory from attackers, which diverts their behavior from initial user intent and finally causes task failure. However,...

https://arxiv.org/abs/2607.28103

00 CONF

$\mathrm{R}{\scriptstyle\mathrm{EAL}}\mathrm{G}{\scriptstyle\mathrm{UARD}}$: Reliable, Efficient, and Auditable LLM Safeguards via Latent Reasoning

ArXiv CS.AI 50%

arXiv:2608.03838v2 Announce Type: replace Abstract: Reasoning-based large language model (LLM) safeguards improve safety classification accuracy by generating explicit rationales before making decisions. However, generating explicit rationales adds substantial decoding overhead...

https://arxiv.org/abs/2608.03838

01 CONF

RetailAgent: Structured Adverse Timing in Self-Conditioned Multimodal LLM Trading Agents

ArXiv CS.AI 50%

arXiv:2608.28399v2 Announce Type: replace Abstract: In financial markets, a sequential policy that reacts systematically to price movements may become predictable to other market participants. This paper studies whether large language model (LLM) agents exhibit such directional...

https://arxiv.org/abs/2608.28399

02 CONF

Right Answer, Wrong Reason: Accuracy, Consistency, and Consensus Are Misleading Indicators of LLM Faithfulness in Clinical Decision Support

ArXiv CS.AI 50%

arXiv:2609.32817v2 Announce Type: replace Abstract: Clinical Large Language Models (LLMs) achieve strong medical-exam accuracy; however, a correct answer does not guarantee that the explanation names the concepts that actually drove the decision. We introduce three lightweight,...

https://arxiv.org/abs/2609.32817

03 CONF

Can Generative AI Automate Data Extraction for Meta-Analysis? A Case Study on Intercropping Research

ArXiv CS.AI 50%

arXiv:2609.35089v2 Announce Type: replace Abstract: Meta-analysis is the synthesis of information from multiple sources to arrive at an overarching conclusion. There is a large need for meta-analysis in agricultural research to synthesize what is known and analyze overarching...

https://arxiv.org/abs/2609.35089

04 CONF

Behavioral History Outperforms Descriptions of the Person for LLM Synthetic Personas

ArXiv CS.AI 50%

arXiv:2610.03998v2 Announce Type: replace Abstract: Large language models (LLMs) are increasingly used as synthetic personas representing survey respondents. Their validity as substitutes for particular respondents depends on whether they reproduce individuals' decisions. We...

https://arxiv.org/abs/2610.03998

05 CONF

MS-Exam-Gen: An Automatic Benchmark Construction Toolkit for Evaluating LLMs on Multiple Sclerosis-MRI Imaging Knowledge

ArXiv CS.AI 50%

arXiv:2610.06170v2 Announce Type: replace Abstract: Biomedical large language model (LLM) evaluation requires auditable assessment of narrow, evolving, source-grounded subspecialty knowledge. Multiple sclerosis MRI (MS-MRI) provides a high-stakes textual-knowledge test case...

https://arxiv.org/abs/2610.06170

06 CONF

Evolving in Thought Space: Training a Small Model at Test Time Unlocks Better Discoveries

ArXiv CS.AI 50%

arXiv:2610.06269v2 Announce Type: replace Abstract: Open-ended scientific discovery often requires repeatedly proposing and evaluating candidate solutions. LLM-based systems can support this process by generating and refining executable solutions from verifier feedback. Methods...

https://arxiv.org/abs/2610.06269

07 CONF

NeuSE: Neural SE(3)-Equivariant Embedding for Long-Term Object-Based Simultaneous Localization and Mapping

ArXiv CS.AI 50%

arXiv:2303.07308v3 Announce Type: replace-cross Abstract: We present NeuSE, a novel Neural SE(3)-Equivariant Embedding for objects, and illustrate how it supports object-based Simultaneous Localization and Mapping (SLAM) for consistent spatial understanding with long-term scene...

https://arxiv.org/abs/2303.07308

08 CONF

ReMAP: Lossless Residual Neural Lifting for Scalable MAP Inference in Arbitrary-Order Markov Random Fields

ArXiv CS.AI 50%

arXiv:2411.18954v5 Announce Type: replace-cross Abstract: Maximum a posteriori inference in a finite Markov random field is generally NP-hard. We study an instance-wise optimizer that minimizes the exact multilinear energy under product marginals. The canonical parameterization...

https://arxiv.org/abs/2411.18954

09 CONF

Open Foundation Models in Healthcare: Challenges, Paradoxes, and Opportunities with GenAI Driven Personalized Prescription

ArXiv CS.AI 50%

arXiv:2502.04356v2 Announce Type: replace-cross Abstract: In response to the success of proprietary Large Language Models (LLMs), there is a growing interest in developing open, non-proprietary LLMs and AI foundation models (AIFMs) for transparent use in academic, scientific,...

https://arxiv.org/abs/2502.04356

10 CONF

Toxicity Is a Property of Communicative Events: A Contextual Stress Measurement Framework for NLP

ArXiv CS.AI 50%

arXiv:2503.16072v5 Announce Type: replace-cross Abstract: Toxicity detection has become core safety infrastructure for online moderation, dataset filtering, and deployed language-model systems. Yet most detectors still treat toxicity as an intrinsic property of isolated text....

https://arxiv.org/abs/2503.16072

11 CONF

Horizon Scans can be accelerated using novel information retrieval and artificial intelligence tools

ArXiv CS.AI 50%

arXiv:2504.01627v2 Announce Type: replace-cross Abstract: Introduction: Horizon scanning in healthcare assesses early signals of innovation, crucial for timely adoption. Current horizon scanning faces challenges in efficient information retrieval and analysis, especially in grey...

https://arxiv.org/abs/2504.01627

12 CONF

Graph Neural Network-Based Reinforcement Learning for Controlling Biological Networks - the GATTACA Framework

ArXiv CS.AI 50%

arXiv:2505.02712v4 Announce Type: replace-cross Abstract: Cellular reprogramming, the artificial transformation of one cell type into another, has been attracting increasing research attention due to its therapeutic potential for complex diseases. However, identifying effective...

https://arxiv.org/abs/2505.02712

13 CONF

Generative AI for Autonomous Driving: A Review

ArXiv CS.AI 50%

arXiv:2505.15863v2 Announce Type: replace-cross Abstract: Generative AI (GenAI) is rapidly advancing the field of Autonomous Driving (AD), extending beyond traditional applications in text, image, and video generation. We explore how generative models can enhance automotive...

https://arxiv.org/abs/2505.15863

14 CONF

Simulating Student-Tutor Interactions with LLMs for Multilingual Math Feedback

ArXiv CS.AI 50%

arXiv:2506.04920v2 Announce Type: replace-cross Abstract: Large language models (LLMs) have demonstrated the ability to generate formative feedback and instructional hints in English, making them increasingly relevant for AI-assisted education. However, their ability to provide...

https://arxiv.org/abs/2506.04920

15 CONF

From Empirical Evaluation to Context-Aware Enhancement: Repairing Regression Errors with LLMs

ArXiv CS.AI 50%

arXiv:2506.13182v4 Announce Type: replace-cross Abstract: [...] Since then, various APR approaches, especially those leveraging the power of large language models (LLMs), have been rapidly developed to fix general software bugs. Unfortunately, the effectiveness of these advanced...

https://arxiv.org/abs/2506.13182

16 CONF

Co-Creative Learning via Metropolis-Hastings Interaction between Humans and AI

ArXiv CS.AI 50%

arXiv:2506.15468v2 Announce Type: replace-cross Abstract: We propose co-creative learning as a novel paradigm where humans and AI, i.e., biological and artificial agents, mutually integrate their partial perceptual information and knowledge to construct shared external...

https://arxiv.org/abs/2506.15468

17 CONF

Physics-Informed Graph Neural Networks to Reconstruct Local Fields Considering Finite Strain Hyperelasticity

ArXiv CS.AI 50%

arXiv:2507.05291v2 Announce Type: replace-cross Abstract: We propose a physics-informed machine learning framework called P-DivGNN to reconstruct local stress fields at the micro-scale, in the context of multi-scale simulation given a periodic micro-structure mesh and mean,...

https://arxiv.org/abs/2507.05291

18 CONF

Unequal Voices: How LLMs Construct Constrained Queer Narratives

ArXiv CS.AI 50%

arXiv:2507.15585v2 Announce Type: replace-cross Abstract: One way social groups are marginalized in discourse is that the narratives told about them often default to a narrow, stereotyped range of topics. In contrast, default groups are allowed the full complexity of human...

https://arxiv.org/abs/2507.15585

19 CONF

Web3 x AI Agents: Landscape, Integrations, and Foundational Challenges

ArXiv CS.AI 50%

arXiv:2508.02773v4 Announce Type: replace-cross Abstract: The convergence of Web3 technologies and AI agents is a rapidly evolving frontier that is reshaping decentralized ecosystems. This paper presents the first comprehensive landscape analysis of the intersection between Web3...

https://arxiv.org/abs/2508.02773

20 CONF

RCR-Router: Efficient Role-Aware Context Routing for Multi-Agent LLM Systems with Structured Memory

ArXiv CS.AI 50%

arXiv:2508.04903v4 Announce Type: replace-cross Abstract: Large language model (LLM) agents are increasingly organized into multi-agent systems for complex reasoning and decision-making tasks. However, scaling such systems can substantially increase inference cost, especially...

https://arxiv.org/abs/2508.04903

21 CONF

SciRerankBench: Benchmarking Rerankers Towards Scientific Retrieval-Augmented Generation with LLMs

ArXiv CS.AI 50%

arXiv:2508.08742v3 Announce Type: replace-cross Abstract: Scientific literature question answering is a pivotal step towards new scientific discoveries. Recently, \textit{two-stage} retrieval-augmented generated large language models (RAG-LLMs) have shown impressive advancements...

https://arxiv.org/abs/2508.08742

22 CONF

Cognifying Education: Mapping AI's transformative role in emotional, creative, and collaborative learning

ArXiv CS.AI 50%

arXiv:2509.25266v2 Announce Type: replace-cross Abstract: Artificial intelligence (AI) is rapidly reshaping educational practice, challenging long held assumptions about teaching and learning. This article integrates conceptual perspectives from recent books (Genesis by Eric...

https://arxiv.org/abs/2509.25266

23 CONF

Energy-Efficient Deep Learning Without Backpropagation: A Rigorous Hardware-Validated Benchmarking Study of Forward-Only Algorithms

ArXiv CS.AI 50%

arXiv:2511.01061v2 Announce Type: replace-cross Abstract: Backpropagation (BP) powers modern deep learning, but it must hold every activation in memory until a global backward sweep consumes it, and that sweep locks each layer until the one above has finished. Forward-only...

https://arxiv.org/abs/2511.01061

24 CONF

Semantics as a Shield: Label Disguise Defense (LDD) against Prompt Injection in LLM Sentiment Classification

ArXiv CS.AI 50%

arXiv:2511.21752v3 Announce Type: replace-cross Abstract: Prompt injection can cause large language models (LLMs) to deviate from their intended tasks by following instructions embedded in untrusted input. We investigate whether changing a classifier's output labels improves...

https://arxiv.org/abs/2511.21752

25 CONF

ECLIPTICA -- A Framework for Switchable LLM Alignment via CITA - Contrastive Instruction-Tuned Alignment

ArXiv CS.AI 50%

arXiv:2601.06157v2 Announce Type: replace-cross Abstract: Alignment in large language models (LLMs) is still largely static: after training, the policy is frozen. DPO, GRPO methods typically imprint one behavior into the weights, leaving little runtime control beyond prompt...

https://arxiv.org/abs/2601.06157

26 CONF

Architecture Before the Formula: Training-Dependent Computational Semantics in Transformers

ArXiv CS.AI 50%

arXiv:2601.11618v4 Announce Type: replace-cross Abstract: An operation's input-output function does not, by itself, determine its computational role within a represented process. That role also depends on the processes producing its inputs and receiving its outputs, a...

https://arxiv.org/abs/2601.11618

27 CONF

NeuroVLA: A Brain-inspired Embodied Intelligence for Fluid and Fast Reflexive Robotics Control

ArXiv CS.AI 50%

arXiv:2601.14628v2 Announce Type: replace-cross Abstract: Vision-language-action (VLA) models inherit powerful semantic representations from large language models, but carrying over their design priorities can leave the demands of physical control under-specified. Stable...

https://arxiv.org/abs/2601.14628

28 CONF

Where is the multimodal goal post? On the Ability of Foundation Models to Recognize Highlights in Videos

ArXiv CS.AI 50%

arXiv:2601.16333v3 Announce Type: replace-cross Abstract: Foundation models are used for many real-world applications involving language understanding and generation from temporally-ordered multimodal events. In this work, we study the ability of models to identify the most...

https://arxiv.org/abs/2601.16333

29 CONF

The pretraining domain outweighs the training objective in setting the privacy-utility trade-off of differentially private medical image analysis

ArXiv CS.AI 50%

arXiv:2601.19618v3 Announce Type: replace-cross Abstract: Differential privacy limits how much each training image can influence a medical imaging model but lowers its diagnostic accuracy. Pretrained weights reduce this loss. It is unknown whether the pretraining objective or...

https://arxiv.org/abs/2601.19618

30 CONF

From Associations to Activations: Comparing Behavioral and Hidden-State Semantic Geometry in LLMs

ArXiv CS.AI 50%

arXiv:2602.00628v3 Announce Type: replace-cross Abstract: We investigate the extent to which an LLM's hidden-state geometry can be recovered from its behavior in psycholinguistic experiments. Across eight instruction-tuned transformer models, we run two experimental paradigms --...

https://arxiv.org/abs/2602.00628

31 CONF

TTT-Parkour: Rapid Test-Time Training for Perceptive Robot Parkour

ArXiv CS.AI 50%

arXiv:2602.02331v2 Announce Type: replace-cross Abstract: Achieving highly dynamic parkour with humanoid robots on complex terrains unseen during the pre-training stage remains a challenge in robotics. Although general locomotion policies demonstrate capabilities across broad...

https://arxiv.org/abs/2602.02331

32 CONF

Prompt Interference in LLM Post-Training: Why Pass@k Optimization Can Degrade Pass@1

ArXiv CS.AI 50%

arXiv:2602.21189v3 Announce Type: replace-cross Abstract: Pass@k is a widely used performance metric for verifiable large language model tasks, including mathematical reasoning, code generation, and short-answer reasoning. It defines success if any of k independently sampled...

https://arxiv.org/abs/2602.21189

33 CONF

OxyGen: Unified KV Cache Management for VLA Inference under Multi-Task Parallelism

ArXiv CS.AI 50%

arXiv:2603.14371v4 Announce Type: replace-cross Abstract: Embodied AI agents increasingly require parallel execution of multiple tasks, such as manipulation, conversation, and memory construction, from shared observations under distinct time constraints. Recent...

https://arxiv.org/abs/2603.14371

34 CONF

PermaFrost-Attack: Stealth Pretraining Seeding(SPS) for planting Logic Landmines During LLM Training

ArXiv CS.AI 50%

arXiv:2604.22117v3 Announce Type: replace-cross Abstract: Aligned large language models (LLMs) remain vulnerable to adversarial manipulation, and their reliance on web-scale pretraining creates a subtle but consequential attack surface. We study Stealth Pretraining Seeding...

https://arxiv.org/abs/2604.22117

35 CONF

How Do Novice Describers Edit AI Drafts? The Role of Draft Quality in Audio Description Authoring

ArXiv CS.AI 50%

arXiv:2605.05348v2 Announce Type: replace-cross Abstract: Audio description (AD) narrates visual elements in video for blind and low-vision audiences. Recent work has shown that giving novice describers an AI-generated draft to start from helps produce higher-quality AD and...

https://arxiv.org/abs/2605.05348

36 CONF

XL-SafetyBench: A Country-Grounded Cross-Cultural Benchmark for LLM Safety and Cultural Sensitivity

ArXiv CS.AI 50%

arXiv:2605.05662v2 Announce Type: replace-cross Abstract: Current LLM safety benchmarks are predominantly English-centric and often rely on translation, failing to capture country-specific harms. Moreover, they rarely evaluate a model's ability to detect culturally embedded...

https://arxiv.org/abs/2605.05662

37 CONF

Beyond Cultural Persona Bias: Response Geometry as a Measure for Language Models and People

ArXiv CS.AI 50%

arXiv:2607.02368v4 Announce Type: replace-cross Abstract: Researchers are increasingly prompting language models with cultural personas so that they answer questionnaires as diverse human groups would. However, studies often report that these persona responses do not predict the...

https://arxiv.org/abs/2607.02368

38 CONF

Conditioned in Training, Dispensable at Inference: Typological Conditioning for Multilingual Speech Translation

ArXiv CS.AI 50%

arXiv:2605.16026v2 Announce Type: replace-cross Abstract: Multilingual speech translation frontends built on speech large language models typically condition on language identity, as a learned embedding or a prompted language name. Such a signal, however, is learned separately...

https://arxiv.org/abs/2605.16026

39 CONF

How Much Can Repeated Running Buy? A Measured Variance Ceiling for LLM Code Evaluation

ArXiv CS.AI 50%

arXiv:2606.00920v2 Announce Type: replace-cross Abstract: A benchmark score is a measurement whose uncertainty has two sources: an aleatoric component from stochastic decoding and an epistemic component from the finite problem set. Using 16 models on 99 recent LeetCode-style...

https://arxiv.org/abs/2606.00920

40 CONF

The Value of Spike Timing: A Leakage-Resistant Benchmark of SNN Design Choices for Network Intrusion Detection

ArXiv CS.AI 50%

arXiv:2606.01442v3 Announce Type: replace-cross Abstract: Spiking neural networks (SNNs) are increasingly studied for network intrusion detection, but comparative evidence on how neuron models and spike encodings affect performance remains limited. Evaluation choices can further...

https://arxiv.org/abs/2606.01442

41 CONF

Improving Heart-Focused Medical Question Answering in LLMs via Variance-Aware Rubric Rewards with GRPO

ArXiv CS.AI 50%

arXiv:2606.05174v2 Announce Type: replace-cross Abstract: Large language models can support medical question answering, but privacy, cost, and reliability still limit their clinical use in focused domains. Can rubric-based reinforcement learning bring a locally deployable model...

https://arxiv.org/abs/2606.05174

42 CONF

LLM Judges Have Dark Current: Stratified Null and Utility Tests for LLM Judge Reliability

ArXiv CS.AI 50%

arXiv:2606.15610v2 Announce Type: replace-cross Abstract: Passing an exact-clone null gate does not entail passing a matched target-selection utility gate. We call a non-tie preference between exact duplicates dark current. On 102 prespecified fresh RewardBench 2 rows, all four...

https://arxiv.org/abs/2606.15610

43 CONF

On the Sparsity-Storage-Accuracy Tradeoff in Parsimoniously Activated Dictionary Learning

ArXiv CS.AI 50%

arXiv:2606.22352v2 Announce Type: replace-cross Abstract: Dictionary learning has long been studied from both optimization and probabilistic perspectives. While formulations with element-wise sparsity regularization (e.g., L1-based sparse coding) admit well-established...

https://arxiv.org/abs/2606.22352

44 CONF

Delegating Before Learning: Where Generative AI Sits in Students' Professional Communication

ArXiv CS.AI 50%

arXiv:2608.28837v2 Announce Type: replace-cross Abstract: We conducted an interview study with twelve students on their use of generative AI in academic communication. Students delegated professional messages to AI most where the pressure to sound professional is highest: email...

https://arxiv.org/abs/2608.28837

45 CONF

HALLMARK: How to Diagnose LLM Citation Verifiers and When to Deploy Them

ArXiv CS.AI 50%

arXiv:2607.18360v2 Announce Type: replace-cross Abstract: GPTZero found 53 NeurIPS 2025 accepted papers with hallucinated citations, and venues now screen submissions with automated reference checkers. Rule- and Large Language Model (LLM)-based verifiers are emerging, but many...

https://arxiv.org/abs/2607.18360

46 CONF

ChannelGuard: From Borrowed Safety to Application-Controlled Defenses in Multi-Agent LLMs

ArXiv CS.AI 50%

arXiv:2607.19430v3 Announce Type: replace-cross Abstract: AI assistants increasingly work as teams, where one model makes a plan, others search the web, use tools, or read stored information, and another writes the final answer. This creates new security risks because harmful...

https://arxiv.org/abs/2607.19430

47 CONF

From Chasing Ghosts to Missed Attacks: Perspectives and Perceptions of SOC Practitioners on LLM Integration, Risks, and Readiness

ArXiv CS.AI 50%

arXiv:2608.00672v2 Announce Type: replace-cross Abstract: Security Operations Centers (SOCs) process large volumes of security events, requiring analysts to accurately detect and assess ongoing cyberattacks under time pressure. Recent advances in Large Language Models (LLMs)...

https://arxiv.org/abs/2608.00672

48 CONF

Lost in Context Compaction: When LLM Systems Drop Side Constraints

ArXiv CS.AI 50%

arXiv:2608.11242v2 Announce Type: replace-cross Abstract: When the context window is under pressure, LLM systems compact prior context to continue ongoing tasks. We study Side Constraints (SCs), user-issued instructions such as "do not delete any emails until I confirm," that...

https://arxiv.org/abs/2608.11242

49 CONF

A, B, C, or Stable? Accuracy and Order Sensitivity Diverge in LLM Multiple-Choice Evaluation

ArXiv CS.AI 50%

arXiv:2608.11947v2 Announce Type: replace-cross Abstract: Multiple-choice question (MCQ) benchmarks are widely used to evaluate large language models, yet a single reported accuracy number is not enough to distinguish a model's latent ability from order sensitivity. Debiasing...

https://arxiv.org/abs/2608.11947

50 CONF

SPINE: Measuring LLM Sycophancy under Sustained Multi-Turn Pressure

ArXiv CS.AI 50%

arXiv:2609.09090v2 Announce Type: replace-cross Abstract: Large language models (LLMs) may abandon correct positions when users push back, exhibiting a failure mode known as sycophancy. Existing evaluations typically use short, prespecified conversations and may therefore miss...

https://arxiv.org/abs/2609.09090

51 CONF

Verification as an Architectural Layer for LLM Agents: A V-Model Design, and a Pilot Study of Its Deterministic Core

ArXiv CS.AI 50%

arXiv:2609.31937v2 Announce Type: replace-cross Abstract: Large language model (LLM) agents built on the ReAct pattern concentrate four responsibilities in one model: selecting a strategy, choosing each action, formatting it, and judging whether the result is adequate. Nothing...

https://arxiv.org/abs/2609.31937

52 CONF

LLM Alignment--Utility Asymmetry under Semantic-Preserving Transformations

ArXiv CS.AI 50%

arXiv:2609.32717v2 Announce Type: replace-cross Abstract: Large Language Model (LLM) alignment is intended to ensure that models remain helpful and safe, but its stability under input distributional shift is not yet fully understood. Prior work shows that aligned models can fail...

https://arxiv.org/abs/2609.32717

53 CONF

Sera: Semantic Representation Aggregation with Knowledge and LLM for Reliable and Interpretable Battery Health Forecasting

ArXiv CS.AI 50%

arXiv:2610.11567v3 Announce Type: replace-cross Abstract: Battery state of health (SoH) forecasting is important for battery management, but remains challenging due to nonlinear degradation and heterogeneity across batteries. Existing data-driven approaches primarily use...

https://arxiv.org/abs/2610.11567